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VirtuHire

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Abstract

Mostly deserving candidates get rejected in this competitive job scenario due to the traditional ways of resume screening. VirtuHire takes this challenge head-on by introducing a live AI interview bot for conducting real-time interviews, providing a more realistic and accurate assessment of potential and skills that an individual may have. VirtuHire not only screens CVs but also conducts interviews as if it were a human, assisting recruiters without replacing them, while automating the hiring process for efficiency.

With support for Urdu, VirtuHire ensures that the recruitment process is accessible and inclusive. The goal is to give every candidate a fair chance to showcase their abilities, regardless of their resume presentation.

The present project offers a web-based solution that utilizes NLP and AI models to conduct technical interviews and automated theoretical tests. In addition to evaluating technical skills, these tools analyze personality traits, cultural fit, and non-verbal cues such as body language, providing recruiters with deeper insights into each candidate. This solution supports the recruitment of candidates from diverse backgrounds, making the hiring process more efficient and inclusive.

Executive Summary

VirtuHire is a project aimed at revolutionizing the traditional recruitment process by leveraging cutting-edge AI technologies to provide a more inclusive and fair candidate evaluation. The platform addresses inherent biases in conventional hiring practices through AI tools such as Natural Language Processing (NLP), Machine Learning (ML), and facial recognition to conduct technical, theoretical, and personality assessments.

Supporting both English and Roman Urdu, the platform fosters inclusivity, ensuring culturally relevant assessments for diverse candidates. By automating key hiring stages, VirtuHire reduces human bias, enhances efficiency, and improves decision-making through data-driven insights. The platform also emphasizes the analysis of personality traits, cultural fit, and non-verbal cues, offering deeper insights into candidates beyond traditional metrics.

Built using the MERN stack, the project integrates AI models via TensorFlow, PyTorch, and NLP tools like spaCy and Hugging Face Transformers. It will be scalable, secure, and user-friendly, providing recruiters with detailed analytics and reports to assist in the hiring process.

The project recognizes the balance between automation and human judgment, ensuring AI assists rather than replaces recruiters. Industry collaboration will guide the platform's development, enhancing its practical impact in real-world recruitment.

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Chapter 1 Introduction

The recruiting process is the most fundamental defining aspect of an organization. Current recruitment approaches have lost sight of addressing the bias and inefficiency that is associated with resume screening and candidate evaluation. In today's competitive corporate world, it's a fairly common for qualified applicants to be overlooked, which can then have a very negative impact on the organisation's human resource and general performance.

VirtuHire aims to address this problem through the use of technologies like Natural Language Processing (NLP), Machine Learning (ML) and facial recognition. This online platform manages a whole range of recruitment activities such as CV screening, conducting technical interviews, and administering personality tests. It also enables the easier and fairer decision making of recruitment processes through VirtuHire as the platform offers an overall assessment of candidates which includes personality, culture, as well as body language. Additionally, the platform supports English and Roman Urdu, promoting inclusivity across diverse candidates.

1.1 Purpose of this Document

This document outlines the goals and scope of our FYP, consisting of both development and research elements. It provides a clear overview of the project, highlighting the use case of the application and approach to tackle current challenges in contemporary recruitment processes.

In this report, we have mentioned our specific objectives, including the integration of essential features and functionalities, as well as our research work on exploring how AI can enhance recruitment processes. By providing a structured framework, we aim to help readers understand the technical outcomes we aspire to achieve.

Additionally, the document covers the tools, technologies, and methodologies we are using to design and develop the system, while also acknowledging any limitations,risks or assumptions. The layout is designed to take readers through the various stages of our development journey, backed by our research work into recruitment practices and technology trends. Furthermore, this document also aims to attract potential investors and companies that may be interested in investing in or acquiring our project.

1.2 Intended Audience

This document serves several important audiences. It is intended for FYP evaluators to assess the project's viability and evaluation, FYP supervisor to provide guidance and feedback throughout the

development process. Developers will reference this document to ensure clarity on the project requirements. Additionally, it aims to attract future investors interested in supporting the platform's growth and innovation. Finally, it serves as a resource for those engaged in future research and studies related to AI and Machine Learning applications in recruitment.

1.3 Definitions, Acronyms, and Abbreviations

FYP: Final Year Project

UI: User Interface

UX: User Experience

ATS: Applicant Tracking Systems

CRUD: Create, Read, Update, Delete

API: Application Programming Interface

SDG: Sustainable Development Goal

GDPR: General Data Protection Regulation.

AI: - Artificial Intelligence

CV - Curriculum Vitae

NLP - Natural Language Processing

LLaMA 3 - Large Language Model Meta AI v3

MBTI - Myers-Briggs Type Indicator

Agile: Agile Development

SCRUM: Scrum Development

1.4 Conclusion

In conclusion, the report structure is as follows: Chapter 2 includes the project vision, presenting VirtuHire and its purpose in shaping the direction of new recruitment practices. Chapter 3 involves a review of pertinent literature and discussion of relevant and existing work in AI-driven recruitment systems, which will help frame our project's unique contributions. Chapter 4 consists of the software requirements and specifications that describe a closer look at the core components and features of the platform. Chapter 5 describes the methodology we will employ, and what tools, techniques, and frameworks are to be used within the application. Finally, Chapter 6 dives into designing at both high and low levels.

Chapter 2 Project Vision

In today's rapidly evolving industry, manual recruitment techniques have started to show their limitations ,they lack efficiency and accuracy and are often biased leading to wrong hiring decisions. Our project, VirtuHire intends to provide an innovative solution that uses modern technologies to streamline the hiring process. By prioritizing efficiency, fairness, and inclusivity, VirtuHire ensures a more comprehensive and unbiased candidate evaluation, ultimately leading to smarter, more effective hiring outcomes.

2.1 Problem Domain Overview

Traditional recruitment processes in the industry are heavily reliant on outdated resume screening methods, which only address specific areas of a resume moreover they tend to focus on semantics and how the resume is written and often fail to determine the true capabilities of an applicant. This dependency on resumes can introduce bias, as candidates with less polished resumes are frequently overlooked, despite having the necessary skills and potential. Additionally, recruiters struggle to process the large numbers of applications received efficiently ,which makes it difficult to assess each resume thoroughly. The recruitment process also lacks the ability to cater non-English speaking applicants.This platform will cater to both recruiters and applicants, providing them a transformative solution to automate and enhance candidate evaluations, ensuring fairness, inclusivity, and efficiency, making VirtuHire an essential tool for modern recruitment.

2.2 Problem Statement

The key issue in the recruitment industry is the sole dependency on resumes and Applicant Tracking Systems (ATS), which often filter out competent candidates based on superficial qualifications. These systems can not evaluate candidates' technical skills, soft skills, or personality traits accurately. Furthermore, in regions like Pakistan, language barrier plays a crucial role, inability to support languages i.e Roman Urdu for the local demographic can prevent talented candidates from being evaluated fairly. As discussed earlier, going through large amounts of resumes without due diligence is not possible nor a very efficient approach in this fast paced era of technology. Hence to cater the problems faced both by recruiters and applicants a proper application is needed to facilitate the recruitment process,which can provide unbiased, skills-based evaluations while catering to linguistic diversity and ensure effective hiring.

2.3 Problem Elaboration

VirtuHire seeks to tackle the major issues facing the current recruitment process by concentrating on the following aspects:

2.3.1 Shortlisting of Candidates

The conventional shortlisting process depends on ATS systems that rank candidates based on keyword matches between resumes and job descriptions. This method overlooks candidates who may lack resume optimization skills but are otherwise qualified. VirtuHire replaces this with an AI-driven approach that evaluates candidates through coding challenges and theoretical assessments, ensuring that applicants are shortlisted based on their skills, not their resumes.

2.3.2 Interview Scheduling

Coordinating interviews between recruiters and candidates often leads to delays, especially when dealing with large volumes of applicants. VirtuHire automates the interview scheduling process, allowing candidates to engage with the AI-driven interview bot as soon as they submit their application. This significantly reduces the time recruiters spend managing interview logistics.

2.3.3 Real-Time Feedback

Candidates often face long waiting periods between interview stages, with little to no feedback on their performance. VirtuHire provides real-time feedback on both technical and personality aspects of the interview, giving candidates immediate insights into their strengths and areas for improvement. This also helps recruiters by providing instant access to candidate performance data.

2.3.4 Personality Evaluation

Most recruitment platforms do not assess soft skills, which are critical to team collaboration and workplace success. VirtuHire integrates personality tests(such as Myers-Briggs Personality Type Indicator) to analyze whether an applicant is right for the organization . This provides recruiters with a more comprehensive understanding of a candidate's personality and interpersonal skills.

2.3.5 Multilingual Support

Traditional systems focus almost exclusively on English, which limits the accessibility of these platforms for non-native speakers. By offering Roman Urdu support, VirtuHire broadens its accessibility,

ensuring that candidates from diverse linguistic backgrounds can participate fully in the recruitment process without language being a barrier.

2.4 Goals and Objectives

We aim to accomplish the following goals and objectives.

2.4.1 User-Friendly Web Application

Design an intuitive web application that makes every single step in recruitment a smooth, enjoyable experience for all people involved - both the recruiters and the candidates. From user-friendly features and a clean interface, towards interactions made to feel as natural as possible, we will thus develop a platform in which it would be easier for everyone involved to successfully find their way through the hiring process.

2.4.2 AI-Powered Live Interviews

Integrate an AI-powered platform to conduct live interviews, enabling a thorough evaluation of both technical skills and soft skills through automated assessments.

2.4.3 Semi-Assisted Interviews

Offer a semi-assisted interview feature that allows recruiters to conduct live interviews while benefiting from AI support. The AI will provide real-time feedback, recommend questions, and assist in evaluating candidates, ensuring a more comprehensive assessment.

2.4.4 Multilingual Support

Provide Natural Language Processing (NLP) capabilities to support both Roman Urdu and English, ensuring broader accessibility for candidates from diverse linguistic backgrounds.

2.4.5 Facial Recognition Technology

Utilize facial recognition technology to assess non-verbal communication cues during interviews, offering deeper insights into candidates' interpersonal and personality traits.

2.4.6 Bias Minimization

Minimize human biases by automating candidate evaluations through standardized, objective metrics, ensuring fairness in the recruitment process.

2.4.7 Interview Analysis

Optimize the analysis of interviews to generate detailed summaries, providing recruiters with valuable insights into both candidates' and interviewers' performance.

2.4.8 Effective Communication

Foster effective interaction between recruiters and candidates, ensuring timely feedback and clear communication regarding the next steps in the recruitment process.

2.4.9 Actionable Feedback

Deliver actionable feedback to candidates after each stage of the interview process, helping them understand their performance and next steps.

2.4.10 Profitability and Sustainability

Aim to develop an application is designed to be both profitable and sustainable, enhancing the hiring process while generating consistent revenue through various monetization strategies, such as subscription models and premium features. By providing substantial value to users and fostering collaborative partnerships with industry stakeholders, the platform aims to cultivate a loyal client base. Furthermore, long-term sustainability is prioritized by ensuring a responsible response to dynamic market changes and incorporating user feedback to drive further innovations, thereby promoting successful operation and utilization within a highly competitive recruitment landscape.

2.5 Project Scope

VirtuHire focuses on developing and deploying a user-friendly, web-based application that aims to automate and optimize the recruitment process. The project includes the following core features:

2.5.1 Core Features

The project includes the following core features:

2.5.1.1 Technical Interviews

Conduct technical interviews using coding challenges and theoretical assessments.

2.5.1.2 Personality and Soft Skills Evaluation

Use facial recognition and personality tests to assess personality traits and soft skills during interviews.

2.5.1.3 Semi-Automated Interviews

Assist recruiters during semi-automated interviews.

2.5.1.4 Natural Language Processing (NLP)

Process interview responses in both Roman Urdu and English using NLP models.

2.5.1.5 Comprehensive Candidate Reports

Provide recruiters with access to detailed reports on candidate performance, including feedback on technical and non-verbal assessments.

2.5.1.6 Automated Candidate Communications

Automate communication with candidates, including effective feedback delivery.

2.5.2 Out of Scope

Hosting or server infrastructure for the application. Mobile application development. Integration with third-party recruitment platforms or external APIs.

2.5.3 Technology Stacks

1. **Frontend:** Next.js with modern styling and user interface libraries.
2. **Backend:** Node.js and Django for the server-side logic.
3. **Database:** MongoDB for storing candidate data, recruiter profiles, and interview results.
4. **Natural Language Processing:** spaCy, NLTK for processing responses.
5. **Machine Learning:** TensorFlow, PyTorch for AI-driven features.
6. **Authentication:** Passport.js or Firebase for user authentication.
7. **Large Language Model:** LLaMA for handling AI-driven tasks.

2.5.4 Datasets and Preprocessing

Different datasets will be used for each feature, collected from various sources. In case of limitations, self-made datasets will be used. Preprocessing will be applied to remove irrelevant information from datasets.

2.5.5 Project Management Methodology

Scrum methodology will be followed for managing project tasks and deliverables.

2.5.6 Deliverables

Web Application: A fully functional application that streamlines the recruitment process. Documentation: Detailed documentation covering literature review, requirement specifications, design, development, implementation, and testing phases.

2.5.7 User Manual

A comprehensive guide for both recruiters and candidates on how to navigate and use the platform effectively.

2.6 Sustainable Development Goal (SDG)

VirtuHire supports SDG 8: Decent Work and Economic Growth by promoting inclusive and fair recruitment practices. The platform ensures that job opportunities are accessible to all candidates, irrespective of their background or linguistic proficiency, thus contributing to economic growth and social equality.

2.7 Constraints

1. Development must be completed within the academic year. This requires efficient use of resources and adherence to deadlines.
2. Ensuring compliance with ethical standards in AI, particularly with facial recognition and data privacy, is crucial.
3. Limited access to high-quality datasets, such as Roman Urdu NLP may affect the accuracy of the system's language processing capabilities.
4. Recruiters following the traditional recruitment process might be reluctant to adopt this system.

2.8 Business Opportunity

VirtuHire has huge business opportunities to tap into with AI recruitment as using it would save companies much time and reduce the cost of hiring while improving the quality of hires because there would now be objective, skills-based evaluations. For corporations, a subscription tier system could be given and for large corporations, contractual agreements can be made. The specific characteristic of this platform is that it supports Roman Urdu. This enables the site to access the relatively untouched Pakistani market, hence giving the company an advantage in locations where language becomes a problem for recruitment.

2.9 Stakeholders Description/ User Characteristics

Identify and briefly describe who the users of this system will be, and their roles.

2.9.1 Stakeholders Summary

1. **System Administrators:** Maintain the system, manage users and user data, and ensure that the platform operates efficiently.
2. **Recruiters:** Use the platform to assess candidates' skills and personality traits.
3. **Candidates:** Apply for jobs, complete interviews, and receive performance feedback through the platform.
4. **Organizations:** Corporations looking for an effective and streamlined recruitment process,be able to buy the application,set a centralized recruitment process.

2.9.2 Key High-Level Goals and Problems of Stakeholders

2.9.2.1 Recruiters

1. **Goal:** Automate the evaluation of both technical and soft skills to streamline the hiring process.
2. **Problem:** Difficulty in filtering high volumes of applicants using traditional CV-based methods, leading to inefficiencies and potential biases in the hiring process.

2.9.2.2 Candidates

1. **Goal:** Receive fair, unbiased, and skill-based assessments that reflect their actual potential, beyond their resumes.

2. **Problem:** Struggles in presenting themselves effectively through resumes, particularly for candidates who may not be native English speakers or are less proficient in resume-building.

2.9.2.3 System Administrators

1. **Goal:** Ensure the platform runs smoothly by managing user data, system security, and overall performance.
2. **Problem:** Managing the system's data integrity and security while maintaining optimal performance as the platform scales with more users and larger datasets.

2.9.2.4 Organizations

1. **Goal**: To enhance the efficiency of recruitment processes that could identify and select the best talent in the shortest time possible, minimizing the time-to-hire.
2. **Problem:** The traditional recruitment ways of today are cumbersome and behind schedule, with factors such as biasness coming in to make it unachievable that the best among students be selected hence hiring mistakes that contribute to poor organizational growth and productivity.

2.10 Conclusion

VirtuHire is a web-based platform designed to revolutionize recruitment by using AI, NLP, and facial recognition to automate and enhance the hiring process. The vision is to create a more efficient, fair, and unbiased system by integrating live AI-driven interviews, technical assessments,multilingual support and personality evaluations. The goal is to provide a user-friendly platform that shifts the focus from traditional resume-based methods to skills and potential. While challenges such as implementing advanced technologies and user training exist, VirtuHire offers immense opportunities to streamline recruitment, improve communication between employers and candidates, and reduce bias in hiring decisions.

Chapter 3 Literature Review / Related Work

This section includes detailed literature review related to the automated recruitment process. We have explored how artificial intelligence is shaping HR management, the best automation practices to get efficient and fair hiring results, and the technologies that boost the automation in hiring.

3.1 Definitions, Acronyms, and Abbreviations

AI: Artificial Intelligence

ML: Machine Learning

LLM: Large Language Models

NLP: Natural Language Processing

MBTI: Myers-Briggs Type Indicator

SVM: Support Vector Machines

HR: Human Resource

RU-BiLSTM: Roman Urdu-Bidirectional Long Short Term Memory

CVs: Curriculum Vitae

SME: Small and Medium Enterprises

e-HRM: Electronic Human Resource Management

HRIS: Human Resource Information System

ATS: Application Tracking System

3.2 Detailed Literature Review

The following sections provide each review material in detail including its summary, the critical analysis, and relation to our work.

3.2.1 Ethics of AI-Enabled Recruiting and Selection: A Review and Research Agenda[1]

As AI is playing a growing role in the recruitment process, fairness, transparency, and accountability in decision-making have become matters of concern. Based on the concerns above, the article reviews the challenges and presents frameworks for organizations to embrace more ethical AI systems. The authors consider how employers, job seekers, and regulatory bodies experience the influence of AI

on recruitment, within which recommendations are made for future research to tackle these ethical dilemmas.

3.2.1.1 Summary of the research item

In this article, the ethical considerations involved in AI-based recruitment are critically examined and the factors of algorithmic bias, fairness, transparency, privacy, and accountability are discussed. The opinions of employers, job applicants, and regulatory authorities are discussed, and a research agenda is suggested for solving the ethical concerns arising from the increase in the use of AI in recruitment.

3.2.1.2 Critical analysis of the research item

The review presents a detailed framework for the reader to appreciate and evaluate in detail the impact of AI on recruitment. A unique aspect of the present paper, which is rare in many papers, is that it addresses the ethical issues related to AI – we consider various stakeholders' concerns. However, the research is indeed very theoretical and fails to dig deep into any of those ethical issues in terms of real implementable recruitment tools. It also does not propose any solutions that can be taken to eliminate or reduce the scope of biases and other transparency challenges in the AI models, consequently making it useless for developers in the near term.

3.2.1.3 Relationship to the proposed research work

This paper is crucial for laying the ethical foundation of our project. The terms of algorithmic justice and transparency will be of paramount importance in the candidate evaluation process of our project's developed system, as many systems do consider such ethical aspects. Most of these ethical frameworks can be adhered to for compliance of our system with industry requirements and expectations of society with respect to the methods used in the evaluation of fairness and accountability.

3.2.2 AI in talent acquisition: A review of AI-applications used in recruitment and selection[2]

AI technology revolutionizes the sourcing of talents by automating many stages of recruitment, ranging from resume screening to candidate assessments. The paper gives an overview of current AI applications used by companies to improve hiring processes. Strengths and weaknesses of using different AI-driven tools are discussed; their contribution to candidate evaluation while hastening up the hiring process shall be explained.

3.2.2.1 Summary of the research item

This article reviews the inclusive collection of AI applications in recruitment and selection including resume screening software to AI usage based interviewing systems. In particular, the review provides insights on the adoption of AI in enhancing candidate evaluation, increasing the speed of hiring, and reducing the length of time taken to fill positions. It also considers the issues encountered in the use of AI in implementing HR systems.

3.2.2.2 Critical analysis of the research item

The review by and large succeeds in achieving the objective of this research by providing a descriptive outlook on the various forms of AI used in recruitment systems. It is about the case studies, whereas there is a lack of coverage of subject response on technical advancement and deployment of these systems. The paper also falls short in addressing what solutions are available for small and medium businesses in order to employ these technologies on a tight budget. While it does a great job charting the waters today, it cannot bring itself to narrow down to numbers or even case studies of how these very tools have helped enhance recruitment or the outcome of the resolution process.

3.2.2.3 Relationship to the proposed research work

It is crucial to carry out this review in order to place our work in the context of the existing AI recruitment tools and their application. Noting the pros and pitfalls of existing applications, we can seek to reinforce our project by addressing shortcomings, which for example include enhancing the capabilities of AI in assessing prospects' profiles beyond the pages of their CVs, to the more complex areas of their personalities and skills sets.

3.2.3 Artificial Intelligence Chatbots are New Recruiters[3]

AI-powered chatbots have emerged as the tool for the automation of the hiring process, and in its initial stage. This paper is thus able to explain the function of a chatbot in terms of candidate interaction and how it has the capacity to automate all schedules for interviews, answer FAQs, and provide feedback. However, the use of these machines brings on its own issues concerning personal interaction and user experience, while easing the workload off human beings and being potentially useful in increasing the engagement of candidates.

3.2.3.1 Summary of the research item

This study deals with the use of AI chatbots to change how the recruitment process is conducted. It explains the importance of chatbots in prospecting candidates, performing administrative functions, replying to candidate FAQs and scheduling interviews, and improving the candidate journey. The benefit of employing chatbots at the initial recruitment stages is also discussed in detail.

3.2.3.2 Critical analysis of the research item

The article makes a convincing argument why it is essential to use chatbots for recruitment including the savings on work and the quicker responses that can be received by the candidates. The study, however, does not go further to examine the limitations, for example, the cold interaction that occurs between the candidates and the recruitment systems with no human being involved or the loss of certain individuals who are not technologically minded and do not want robots/call centers speaking on behalf of the recruiting authority. To begin with, this paper lacks concrete evidence through numbers that would help show the value that such a practice could bring in recruitment in quantitative terms.

3.2.3.3 Relationship to the proposed research work

The cited study may be used a great deal by our project in terms of using chatbots or other automated systems in the interaction with aspiring candidates. Automating low-level tasks like organizing schedules and dealing with frequently asked questions can reduce the workload on the structure for the most important stages such as the interviews. This paper advocates the notion that systems of recruiting using AI chatbots will indeed increase the productivity of recruitment systems and will therefore be ideal for VirtuHire.

3.2.4 Ethics and discrimination in artificial intelligence-enabled recruitment practices[4]

As the adoption of AI recruitment systems increases, numerous debates over discrimination and ethical usage arise. First, this paper brings forth the dangers of bias and biased automated decision making and especially avers a regulatory framework that would lead to fairness in AI systems. The study emphasizes the fact that AI systems must never be opaque to avoid ”second-order” unintended racial, gender, or class-based discrimination.

3.2.4.1 Summary of the research item

The present paper focuses on the ethical concerns as well as possible discrimination risks, which accompany AI-enabled recruitment systems. This helps to demonstrate, for instance, how emergency harm

policies can help in addressing risks associated with automated decision-making that perpetuates biases regarding race, gender, or economic status. The paper makes recommendations for more regulations and standards so as to maintain fairness in the control systems of hiring by means of automation.

3.2.4.2 Critical analysis of the research item

The study examines the ethical as well as discrimination risks of the AI recruitment tools in a fairly broad manner. Loss of the provision of proposed regulations on the administration of the AI systems particularly underscores the desire for independent verification of such systems. Though the sobering realities criticizes the gaps in design with the modern AI system, there are very few ways outlined in the design concepts to achieve that. In addition, it does not include a case study which is necessary in demonstrating how the mentioned ethical principles would actually be employed in practice.

3.2.4.3 Relationship to the proposed research work

Addressing potential discrimination risks at the outset will benefit our project by enhancing the credibility of recruitment using AI considering that VirtuHire is planning to incorporate AI to manage technical and personality tests. This research is likely to supplement the course of our work by integrating the ethical principles suggested. As AI applications in recruiting have many potential biases, this research shows how to integrate ethical considerations into the workings of any AI system.

3.2.5 RecruitPro: A Pretrained Language Model with Skill-Aware Prompt Learning for Intelligent Recruitment[5]

This paper examines the opportunities of advanced AI techniques in recruiting, mainly keeping its focus on skill-aware prompt learning. Through advanced AI techniques, there is adaptation of machine learning models to relate candidate skills more directly to job roles and to make the hiring process more efficient and data-driven. RecruitPro is the benchmark in AI-driven recruitment solutions with its introduction of NLP-based frameworks that streamline job matching and skills assessment.

3.2.5.1 Summary of the research item

RecruitPro takes recruitment to a new level by introducing new approaches, in this case, adopting skill-aware prompt learning, which is less likely to be restricted by the domain of application. The model improves the candidate-job fit by appreciating the complex interactions between the job requirements as presented in the job description and the skills of the candidates. The authors designed 13 tasks to examine and evaluate different aspects of the recruitment process, including job classification and skills mismatch.

3.2.5.2 Critical analysis of the research item

This is yet another innovative aspect of RecruitPro that emphasizes on flexibility and adaptability in recruitment and in doing so, responds to the challenges that are associated with the application of machine learning tools stimulated by specific problems. Nevertheless, the constraining character of number and scope of tasks that are predefined and labeled, in this case 13 tasks, limits the system's applicability in real life settings which are almost never static and recruitment needs can change at any time. Also, though the system performs well in relevant task environments, an assessment of how well the system performs in differing recruitment environments is relatively limited. Rather, the system like most beneficiaries in other real life settings would have been able to relate and make much more practical experiences outside the four walls of the created environment.

3.2.5.3 Relationship to the proposed research work

Certain features of RecruitPro's skill-aware prompt learning embed well into our project of assessing the skills and personalities of candidates. We can utilize its framework in boosting the NLP aspects of our system. The skill-based prompt learning method can help improve resume scanning and interview stages by increasing the probabilities of choosing the right candidates. In addition to that, thematic analysis outlined in RecruitPro on benchmarking tasks serves as an advisable strategy in testing how effective is our AI at varied stages of the recruiting process including but not limited to matching candidates to vacancies and evaluating candidates.

3.2.6 Resume Classification Using ML Techniques[6]

The automation of the recruitment process, especially at the screening phase using resume screening, is increasingly being carried out with machine learning models. Evaluation of several ML techniques for the classification of resumes, including decision trees, SVM, and neural networks, with an aim to provide insight into the benefits of automating this stage of the recruitment process, is discussed in this paper. The paper deals with the topic of using ML models to improve recruitment efficiency by reducing manual intervention and increasing accuracy in resume screening.

3.2.6.1 Summary of the research item

This paper addresses the automatic classification of resumes using computer-based decision-making tools. The classification involves machine learning techniques such as decision trees, support vector machines (SVM), and neural networks. The objective is to enhance efficiency in the candidate filtering process. The resume sorting stage is automated to enable rapid processing and reduce the time needed

for reviewing submitted documents, which traditionally relied on extensive manual input.

3.2.6.2 Critical analysis of the research item

The contribution is a partial assurance about the practicality of AI enabled recruitment by quantifying the performance of various ML techniques for resume classification analysis. The model is novel in that it shows how the processes of resume sorting can be improved, time consumed and human discretion minimized. However, such advancement is not replicated when reversal of the process to other recruitment phases like, interview analysis or personality assessment is concerned. Also, there is a possibility of these models performing differently or being affected by the quality and format of the resumes that are input into the system, leading to an irregularly used data.

3.2.6.3 Relationship to the proposed research work

This project will be needing this research during the first stage of our project, where resumes have to be sorted quickly using some screening criteria. We can use one more classification for the same purpose to short list candidates for the next rounds of the recruitment process. Adopting some of the techniques introduced in this research enables our system to automate the early parts of the recruitment process leaving the human resource personnel to deal with the more advanced aspects of recruitment such as interviews and interactions with candidates. Moreover, this paper makes a solid argument for the use of ML models in parsing resumes which can be built on in our AI approach using NLP and psychometric evaluations.

3.2.7 Bias in Multimodal AI: Testbed for Fair Automatic Recruitment[7]

Multimodal AI systems in recruitment have promised so much but have come under fire on the issues of fairness and biases. This paper delves into the challenges of using multimodal AI systems in recruitment by assessing the possible bias in them, especially by race and gender. The research develops a testbed for a rigorous method of assessing how different AI components within recruitment-employing AI, such as facial recognition and NLP-can work to either reduce biases or exacerbate them in decision-making during recruitment.

3.2.7.1 Summary of the research item

This paper exposes the weaknesses embedded in the recruitment systems featuring the multimodal AI, focusing mostly on biases such as race or gender based bias found in automatic recruitment systems. The study is based on a multimodal testbed, one that has been designed to state fairness across and within media types and tools including facial recognition, NLP, candidate assessment tools, and others.

The purpose is to design recruitment systems that allow hiring of people but do so selectively with an emphasis on minimizing the biases associated with recruitment.

3.2.7.2 Critical analysis of the research item

That particular study tackles the probably most critical issue in the domain of AI recruitment – namely bias. The strength of this research lies in the use of a multimodal approach that allows assessment of bias across different methods within the system for example textual analysis and facial recognition. However, the testbed is limited in that it solely focuses on race and gender stereotypes while age and disability biases are not comprehensively targeted. But due to resource constraints, the testbed does not have real trial data which would be the best metric to try and highlight what these fairness measures actually achieve in practice. The paper could have benefited from addressing the integration of the system into practice through industry, which would demonstrate the performance of the system in the current recruitment processes.

3.2.7.3 Relationship to the proposed research work

Having an ethical perspective in the strategy of building the recruitment system in this project. Incorporating some of the bias detection approaches described in this research, our system can reduce the risks of biased decision making especially with respect to hiring policies that involve NLP and facial recognition. In addition, conducting such a testbed in order to verify that our system's fairness can be maintained across various communicative channels may enhance its reliability and the trust of the public on it.

3.2.8 Does AI Debias Recruitment? Race, Gender, and AI's "Eradication of Difference"[8]

This paper explores whether AI recruitment systems actually eliminate prejudice or, in contrast, reinforce existing inequalities. It discusses a range of AI systems and examines their ability to correct matters pertaining to race and gender biases in the recruiting process. The discussion presents some challenges to debiasing AI and promises new avenues for improving those shortcomings.

3.2.8.1 Summary of the research item

In this paper, it is attempted to evaluate the effectiveness of AI technology not only regarding the attempts to reduce the bias in hiring processes but also regarding whether such technologies can help improve the discrepancies present within society. Several case studies are examined in which AI type tools were applied in the process of recruitment and efficacy in minimization of discrimination in hiring was assessed.

3.2.8.2 Critical analysis of the research item

The paper successfully disproves the misconceptions concerning the impartial nature of AI. It substantiates the claim that the majority of such AI based system designs do not revamp bias in the recruitment process as most of them are trained using biased data sets. At the affirmative high point of the claim for better collection and validation processes in the paper, the position fails to provide a concrete method on how these improvements can be implemented in other industries. Its focus on race and gender may be broadened as such data might help in addressing other aspects of diversity like age or disability.

3.2.8.3 Relationship to the proposed research work

It is enough to use different systems and use continual override of discrimination activities to get different systems from any system that can bring such discriminatory conclusions. Adopting methods from this paper will help guarantee that the recruitment system in our organization does not simply aim at reducing the existing bias but rather assists in advancing the recruitment of the right candidate.

3.2.9 Attention-Based RU-BiLSTM Sentiment Analysis Model for Roman Urdu[9]

This paper attempts to present the challenges related to Roman Urdu text sentiment analysis in the advent of non-standardized text inputs in modern communication. An NLP model that can deeply process and analyze any non-English text efficiently is presented in the paper with deep learning techniques. While this model is primarily focused on the task of sentiment analysis, the underlying techniques could be applied to more extensive language processing tasks in recruitment systems.

3.2.9.1 Summary of the research item

This study demonstrates the use of attention based RU-BiLSTM model to conduct sentiment analysis on the Roman Urdu text, a language commonly used in Pakistan. The essay under review highlights the issues of unstructured linguistic data handling and illustrates the ways in which one can effectively employ deep learning to lessen the tension associated with more structured formal languages.

3.2.9.2 Critical analysis of the research item

The ability of the model to effectively analyze a non standardized language such as Roman Urdu is commendable. Nonetheless, its sheer concentration on sentiment analysis and language processing makes it unfit for wider related recruitment problems. Also, the model's performance is largely reliant on good quality data, and not all situations in recruitment will have such data as standard.

3.2.9.3 Relationship to the proposed research work

Although this section is not strictly about recruitment, this paper's concentration on NLP and sentiment analysis also offers some insights for the manipulation of uncontrolled text within the scope of the candidate's responses, especially for the interview analysis part of our project. Similar studies should help improve evaluation of candidates' written skills in several languages, hence extending the recruitment process.

3.2.10 Fairness in AI-Driven Recruitment: Challenges, Metrics, Methods, and Future Directions[10]

There is growing interest in fairness in AI-driven recruitment given the increased employment of such systems. In this paper, an overview of the challenges faced by AI models in regards to attaining fairness, especially as regards biased outcomes, is drawn along with some metrics and methods used to review and enhance improvement in the enhancement of fairness within recruitment systems, and future directions for AI are discussed.

3.2.10.1 Summary of the research item

This paper focuses on the implications of fairness in the processes and outcomes of the use of AI-based recruitment systems. This highlights the risks and the biases in the AI models and recruitment algorithms and how such biases may lead to discrimination against a number of factors such as race, gender, or social class. The authors argue and suggest several ways in which fairness can be assessed or implemented, such as through the use of fairness constraints and adversarial debiasing/fairness-aware ML techniques. They also delve into the ethical aspects of AI use in recruitment and outline possible approaches that can be taken for the construction of fairer AI systems in the future.

3.2.10.2 Critical analysis of the research item

This paper discusses AI recruitment and examines the fairness criteria, identifying the need to include fairness measures for AI system designs. It provides practical fairness evaluation approaches and proposes integration of the approaches in practice, which remain a bit thin on the ground. Additionally, tackling technical fairness only is insufficient given the broader social and ethical concerns of using AI in making hiring decisions, which raises concerns over fairness.

3.2.10.3 Relationship to the proposed research work

This is a very important study for VirtuHire, because it tackles the problem of fairness in AI-powered recruitment. In many ways, we can use the fairness metrics raised in this paper to evaluate and counter bias in the AI models under our control so that none of the recruitment tools we have developed will end up biasing against certain candidates. It is certain that these fairness checks must be done and will even enhance the ethical morality of the project.

3.2.11 Resumate: A Prototype to Enhance Recruitment Process with NLP-based Resume Parsing[11]

Resume screening is one aspect most likely to make an automation process in recruitment more efficient. This paper portrays an NLP-based system that is used for the automation of resume parsing named Resumate. The system retrieves relevant resume information based on natural language processing. Consequently, it filters applicants with the help of HR professionals and thus accelerates the whole recruitment process.

3.2.11.1 Summary of the research item

This paper presents Resumate – a prototype for streamlining the recruitment process through resume parsing. The system is employed to retrieve the relevant information of applicants from their resumes, which allows categorization of the participants' responses according to the job opening. NLP makes it possible to parse the resume and increases the accuracy of hiring while considerably decreasing the time spent screening applicants through the traditional mode of going through the resumes.

3.2.11.2 Critical analysis of the research item

The main advantage of this paper is the practical orientation in solving one of the key problems related to recruitment – resume parsing. Employing NLP techniques for resume parsing and screening applications is especially relevant in high-volume, fast-turnaround recruitment. Nevertheless, 'shallow' academic papers tend to cover only the super technical elements of resume parsing without attempting to examine practical deployment of such automation within the recruitment process (interview scheduling, personality evaluation, etc.). Although NLP aids in resolving some problems like the usability of the information, appropriate resume parsing may still be defeated in situations where candidate documents are not presented in a standardized format.

3.2.11.3 Relationship to the proposed research work

This work has direct relevance to the resume parsing feature of VirtuHire. We may use the NLP approaches proposed in this paper to improve our system's performance in the extraction and analysis of information related to candidates from their resumes. Also, the experience gained from Resumate's prototype can assist us in refining the resume ranking algorithms in our system to ensure it remains both effective and efficient in the screening of candidates.

3.2.12 Role of Information Technology in Human Resource Management of SME: A Study on the Use of Applicant Tracking System[12]

This paper discusses how the existence of Applicant Tracking Systems (ATS) tends to impact SMEs in managing their recruitment processes. The paper addresses the potential for IT-driven HR systems, ATS in particular, to streamline SME recruitment. Centralizing candidate data allows small-to-medium enterprises to increase hiring efficiency and also reaches some of the challenges SMEs may incur while employing such technologies.

3.2.12.1 Summary of the research item

This paper focuses on the extent to which small and medium enterprises (SMEs) utilize Applicant Tracking Systems (ATS) in their recruitment processes. It demonstrates how information technology impacts the human resource function, with a focus on the adoption of ATS by small and medium enterprises. The research explains the advantages of ATS implementation, such as a reduction in time to fill positions, an increase in the number of candidates, and a shortening of the entire recruitment process. It also considers in detail the issues SMEs encounter with such technologies, such as cost, complexity, and lack of user readiness.

3.2.12.2 Critical analysis of the research item

Complementary procedures have been outlined in the paper to show how other SMEs with scarce resources can take advantage of ATS to enhance their recruitment processes. However, its emphasis on the application of ATS technology in SMEs may limit its use in larger companies or in complex recruitment scenarios. The paper also points to concerns about the adoption of cold feet, whereby cost and lack of technical resources are identified as deterrents, but no explicit solutions to counter them are provided.

3.2.12.3 Relationship to the proposed research work

This research supports the broader context of our project by showing the practical aspects of ATS in recruitment. This article will help persuade the automation of recruitment and the centralization of recruitment systems for companies of all sizes. Additionally, understanding why SMEs do not adopt ATS systems will enable us to design VirtuHire to be more user-friendly and scalable for both small and large companies.

3.2.13 HireVue[13]

Video interviews are now in vogue in recruitment with the aid of AI. HireVue is a leading platform integrating AI in interviews via facial expression analysis, intonation in voice, and word usage. This system aims to improve recruitment by analyzing a candidate's thoughts during the interview process, thereby facilitating good hires for an organization.

3.2.13.1 Summary of the research item

HireVue is a video interviewing and assessment platform, utilizing AI in the recruitment process for its customers. It employs AI to measure body language, intonation, and choice of words to determine a candidate's hiring potential. It also assesses applicants' programming skills through coding tests and evaluates their cognitive abilities with psychometric tests. The goal of HireVue is to reduce the manual handling of recruitment processes, such as resume and interview reviews, which can lead to time wastage and discrimination.

3.2.13.2 Critical analysis of the research item

One notable aspect of HireVue is that it can assess candidates across multiple factors using video-based assessments. It has been widely adopted by large organizations, making it recognized as a leader in the AI recruitment market. However, excessive reliance on video clips has brought criticism regarding biases in the system, such as those based on race and gender. Some critics argue that using biased AI simply augments inequality. Many candidates also raise privacy concerns, especially in interviews, since they are unsure of how their data is being used or analyzed.

3.2.13.3 Relationship to the proposed research work

HireVue's facial and speech emotion recognition systems are relevant to the virtual hiring systems conducted in-house at Pants Technologies. Real-time AI interviews can be conducted, addressing some of HireVue's concerns about ethics, education, and data usage without being overly aggressive. VirtuHire

can design or adopt techniques to reduce bias and improve candidate assessment, giving the platform a competitive edge in the market.

3.2.14 Pymetrics[14]

Pymetrics analyzes candidates' emotional and cognitive features based on neuroscience games and AI algorithms, with a focus on eliminating bias in recruitment. This tool evaluates candidates' personality traits and maps them to job roles that match their strengths, enabling more intelligent and fair deployment of individuals through data-driven insights, rather than traditional resume-based hiring practices.

3.2.14.1 Summary of the research item

The Pymetrics platform combines video-game-like applications with AI to assess candidate abilities based on cognitive biases. These games evaluate features such as detail-oriented thinking, risk-taking, decision-making, and problem-solving abilities. By seeking diverse data, the platform's AI attempts to place each candidate into a role that matches their profile, thus overcoming bias. Pymetrics seeks to level the playing field in recruitment by moving away from traditional resume filtering and focusing on individual potential.

3.2.14.2 Critical analysis of the research item

Pymetrics is redefining recruitment engagement, using psychometric gaming instead of traditional CVs to assess candidates, while providing employers with useful data about candidates' cognitive and emotional skills. This approach is especially appealing in industries that value inclusivity. However, challenges remain in applying the system across various industries and job types. Furthermore, the gamified tests may not be applicable to all occupations, and the use of AI raises concerns about data security and the accuracy of algorithms.

3.2.14.3 Relationship to the proposed research work

This research is extremely relevant to VirtuHire because of Pymetrics' use of AI in the recruitment process through cognitive and emotional assessments. VirtuHire can adopt similar approaches by leveraging games for pre-employment screening or testing scenarios to evaluate how well candidates align with specific job roles. Additionally, as VirtuHire aims to ensure fairness in recruitment, applying unbiased AI methods will help to reduce disparities in the hiring process.

3.2.15 Zappyhire[15]

Zappyhire provides an all-rounded AI-driven recruitment automation platform that streamlines hiring from resume screening to onboarding candidates. With features like AI-driven candidate matching, predictive analytics, and automated communication tools, Zappyhire offers reduced recruitment time and a better candidate experience to be placed suitably for fast-paced hiring environments.

3.2.15.1 Summary of the research item

Zappyhire is an AI recruitment platform that automates the entire recruitment process right from candidate sourcing to their onboarding. Some features include automated resume evaluation, AI-based interviewing, and candidate tracking. Zappyhire also ingests resumes and shortlists candidates with the help of AI analytics and assists in strategic decision-making through predictive insights. The platform further aims to enhance the candidate's journey by facilitating faster responses, built-in feedback, and automated updates regarding their applications.

3.2.15.2 Critical analysis of the research item

Judging solely in terms of automation, Zappyhire is proficient since it knows how to provide recruitment workflow from end to end. Such an AI-based resume screening also helps a lot where 'time to hire' matters due to the nature of employment being sought after. On the contrary, this platform falls short at a further stage where the candidate is actually enlisted, which means that no deeper psycho-emotional particle assessment including personality assessment and IQ is currently available. It should also be noted that Zappyhire's predictive models are primarily based on the historical recruitment patterns and therefore they risk being biased towards certain attributes or aspects of a position.

3.2.15.3 Relationship to the proposed research work

The existing capability of Zappyhire to automate the stages of resume sifting out and candidate engagement can be taken into the practices of resume mining and filtering of the initial candidates in VirtuHire. However, our project can further stand out by not only taking what Zappyhire has on predictive modelling and moving beyond it by introducing advanced AI techniques such as personality and technical evaluations of the candidates which will help expand the scope of candidate assessment online.

3.2.16 Glider.ai[16]

Glider.ai uses technical recruitment through automation in its platform, offering an automated coding assessment and real-world skill evaluation. Glider.ai helps employers measure the candidate's techni-

cal capability through automated coding tests and video interviews while keeping the integrity of the assessment intact with fraud-detection mechanisms.

3.2.16.1 Summary of the research item

Glider.ai is a recruitment engine for the assessment of candidates by outsourcing coding tests, video interviews and AI evaluations to the platform. The platform focuses on practical skill testing so the prospective employees can see how they would perform in various tasks relevant to their responsibilities. Furthermore, Glider.ai implements different persuadable fraud mechanisms to monitor the security of remote assessments as well as interpret the activity of the online candidate.

3.2.16.2 Critical analysis of the research item

Glider.ai is unique as it does not depend on CVs or interviews, like traditional methods, for its evaluation but relies on practical approaches. Its automated coding tests are particularly useful in technical job positions that require hands-on skills and experience rather than just attainment of degrees. However, the performance of the candidates with regard to the different activities may often be clouded on the proper evaluation of the soft skills such as interaction and collaboration which so many industries adopt. Also, the readiness of the industry towards the anti-fraud measures against posing and cheating of the examination has their dark sides that should be rectified.

3.2.16.3 Relationship to the proposed research work

Glider.ai's concept of skill-based evaluations could be embedded in the technical component of VirtuHire's offering, which includes conducting coding tests and assessments of problem-solving abilities in the course of interviewing. This is consistent with our aim to evaluate technical skills via real life activities on the platform, but VirtuHire can set itself further apart by offering personality assessment and interviewing alongside the technical evaluations.

3.2.17 Eightfold.ai[17]

Eightfold.ai is AI-powered talent acquisition, predictive analytics hiring solution that has been tailored to match the right candidate with the right job role based on their skills and experience. It can help employers in recruitment efficiency and making more informed hiring decisions to source and skill across different talent pools.

3.2.17.1 Summary of the research item

Eightfold.ai is a hiring solution that integrates artificial intelligence with machine learning to enhance the recruitment process. It has functions to search for candidates, test skills, and offer candidates based on prediction modeling. The system adopts the strategy of enhancing the efficiency and effectiveness of organizations by utilizing the skills and experience of the candidates in filling the gaps available in the organization through the systems deployed more accurately.

3.2.17.2 Critical analysis of the research item

Eightfold.ai is well conformed with its clear definition of its innovative solutions. Emphasis is not only placed on sourcing for candidates, but on deploying talent strategies that augment employee sourcing and employee progression analysis. The platform's focus on skills rather than just matching job titles to resumes may be one of its strengths, as companies tend to look for a broader fit still based on skills. But the downside of this is that the platform might still involve a lot of complexity in the processes as it has to interface with other HR systems that are already in place. It is alarming suggesting racism discrimination especially with artificial intelligence, as much as Eightfold is advocating for diversity there will also be some challenges about bias of AI algorithms if this is not properly taken care of.

3.2.17.3 Relationship to the proposed research work

As is evident Eighfold.ai would be beneficial to VirtuHire since the company plans to use AI to assess the skills and analyze the behaviors of individuals along the life cycle. Learners can use its logic-based candidate recommendation system to incorporate similar approaches in their recruitment system. Furthermore, VirtuHire should focus on ethical concerns in designing AI systems in recruitment processes thus enhancing willingness among job seekers to trust the platform.

3.2.18 Ezyhire[18]

Ezyhire offers preemployment assessment tools that are aimed to evaluate candidates on various dimensions, ranging from technical skills, soft skills, and situational judgment. Being a product that adopts the automation of the assessment process, Ezyhire provides employers with data-based insights into whether candidates would be suitable for specific roles or not, making the recruiting process more streamlined.

3.2.18.1 Summary of the research item

Ezyhire is a better way to be hired assessment tool designed to facilitate the recruitment process by providing multiple tests and assessments before onboarding. The platform equips employers with different

ways of assessing their candidates including coding assessments, personality tests, and situational judgment tests. Ezyhire seeks to enhance the recruitment process effectiveness by offering evidence about the capability as well as the job suitability of the candidates.

3.2.18.2 Critical analysis of the research item

Ezyhire can boast about its unique feature which is the variety of assessments that it provides to evaluate the technical areas and soft skills, which helps to judge candidate's abilities effectively. At the same time, both employers and applicants found it very convenient to carry out the assessment as the interface was user friendly. Still, it seems that it isn't providing such a strong focus on AI based analytics as few of its competitors hence hindering some of the predictive elements of candidate assessment. Above all, even if it is said to have considerable knowledge of the issue, the actual use of its tests does not have a very extensive base.

3.2.18.3 Relationship to the proposed research work

The assessment tools that were used by Ezyhire are reasonably appropriate to the working methods of VirtuHire when trying to assess the skills and personality of candidates. By using similar assessment tools that are designed by Ezyhire, our site will be able to better understand how strong skillets each candidate possesses. Also, concentrating on user experience during the online assessments would increase candidate participation and satisfaction with the entire hiring process.

3.2.18.4 Codility[19]

Codility is a technical hiring platform, majorly offering services through automated coding assessments. The use of the platform leads to companies providing multiple coding challenges that tend to reflect real scenarios. This way, it helps the companies assess the candidate's practical coding skills. In essence, this has helped companies to improve their technical recruitment process through objective evaluation by candidates of problem-solving and coding skills.

3.2.18.5 Summary of the research item

Codility is a web-based service that provides companies with a means to test software developers by employing automated coding challenges. The users of the platform present a catalog of coding tasks and practical situations that can be performed by the candidates within a specified timeframe. The essence of Codility is in making it easy for employers to conduct a technical interview and allow the candidate's coding skills to be tested independently of many factors.

3.2.18.6 Critical analysis of the research item

Codility has a firm reputation for providing a great number of coding games that reflect the abilities and challenges of actual developers working in the field. This enables set up and administration of the tests in a simple straightforward way and the feedback provided to the candidates is instant adding value to their satisfaction. Most of the participants want to attempt just that, standard coding challenge typical of Codility. However, one of the drawbacks is that although Codility is mainly a test of programming ability, it lacks diversity in the means of assessing how candidates think and address challenges. Some candidates may however take time off regular programming/sitting to prepare for Codility tests which affect results where Codility is the only method applied.

3.2.18.7 Relationship to the proposed research work

VirtuHire features a technical evaluation of the candidate, which is why the model adopted by Codility is quite useful for our platform as well. Codility exists in order to allow other coders to screen candidates effectively, therefore in case we also extend their reach, it guarantees that relevant tasks are given to the candidates. Moreover, alongside coding tasks, following the core principles of those personality and behavioral tests.

3.3 Literature Review Summary Table

The following tables 3.1 ,3.2 contains summary of the literature and related application review.

Table 3.1: This table contains summary of the literature review done.

Source	Features	Relevance to VirtuHire	Limitations
Ethics of AI-Enabled Recruiting and Selection: A Review and Research Agenda[1]	Surveys the ethical issues surrounding AI recruiting including just and accurate use of data	Provides a moral base for the use of AI in recruitment processes	Theoretical; does not include practical means of implementing the ethics.
AI in talent acquisition: a review of AI-applications used in recruitment and selection[2]	A growing collection of literature examining numerous AI-based functionalities embedded in HR processes	Illustrates the status of VirtuHire against AI-powered recruitment processes	Lacks detailed discussions on how the technology will be applied and measured.
Artificial Intelligence Chatbots are New Recruiters[3]	Utilizes chatbots for candidate engagement in recruitment processes	Considers chatbot effectiveness for candidate engagement in VirtuHire	Some candidates may find interactions cold; lacks ROI assessment.
Ethics and Discrimination in AI-Enabled Recruitment Practices[4]	Examines ethical issues and discrimination in AI recruitment	Supports ethical principles against bias in VirtuHire	No tangible measures to redress the risks; lacks case studies.
RecruitPro: A Pretrained Language Model with Skill-Aware Prompt Learning for Intelligent Recruitment[5]	Skill-aware prompt articulation towards recruitment policy	Improves NLP in evaluating candidates and matching their skills	Has shortcomings in scalability when deployed in real-life scenarios, has not been widely tried out in many situations.
Resume Classification Using ML Techniques[6]	Automation of resumes classification through the application of ML techniques namely SVM and neural networks	Helps in proper screening of applicants' resumes	Innovates only on the classification of resumes without integration into other recruitment activities.
Bias in Multimodal AI: Testbed for Fair Automatic Recruitment[7]	Assesses the biases seen in AI recruitment settings including gender and race	Offers procedures for reduction of bias in the VirtuHire artificial intelligence models	Investigates biases only; does not explore geometric type bias and has no practical application.

Table 3.2: This table contains summary of the literature review done.

Source	Features	Relevance to VirtuHire	Limitations
Does AI Debias Recruitment? Gender, Race, and AI Eliminating Difference[8]	Investigates biases in recruitment through AI	Critical for VirtuHire in improving recruitment processes	Primarily focuses on race and gender bias; limited scope for other dimensions.
Attention-Based RU-BiLSTM Sentiment Analysis Model for Roman Urdu[9]	Provides insights for managing candidate communication with employers	Offers useful tips for communication strategy	Focused on sentiment analysis; does not extend to broader recruitment processes.
Fairness in AI Driven Recruitment Processes: Barriers and Facilitators[10]	Identifies challenges and fairness metrics in AI recruitment	Useful for evaluating bias in VirtuHire's AI processes	Theoretical notions; practical application is still under-researched.
Resumate: A Prototype to Enhance Recruitment Process with NLP based Resume Parsing[11]	Describes NLP techniques for automating recruitment processes	Enhances resume parsing capabilities of VirtuHire for efficiency	Primarily focused on technology without integrating into larger processes.
Role of Information Technology in Human Resource Management of SME[12]	Studies the use of Applicant Tracking Systems in recruitment for SMEs	Facilitates automation in VirtuHire's processes	Limited validity as it only targets SMEs.

3.3.1 Application Review Summary Table

Table 3.3: This summary table includes various applications relevant to the recruitment process and identifies each column clearly.Relevant applications are presented.

Application	Features	Relevance	Limitations
HireVue[13]	Video interviewing, AI arguments binarized by emotional speech, analysis of intonation	Utilizes AI for real-time interview processes, increasing diversity of applicants	Bias in analytics and concerns regarding candidate confidentiality.
Pymetrics[14]	eLearning apps centered on emotional and cognitive testing; data-based allocation of candidates to positions	Helps to match customers' personalities with job requirements	Limited penetration across diverse roles; data privacy risks.
Zappyhire[15]	Resume parsing automation, candidate management by AI, predictive analytics	Facilitates timely communication and resume flow within VirtuHire	Limited to resumes; does not incorporate other assessments.
Glider.ai[16]	Coding assessments, video interviews, practical exams, anti-scam measures	Ensures candidates are accurately represented through technical evaluation	May overlook soft skills and infrastructure considerations.
Eightfold.ai[17]	Predictive analytics, matching skills to positions, sourcing for diverse candidates	Provides approaches for job and candidate matching to achieve VirtuHire's goals	Risks of over-investment in high-tech solutions for small organizations.
Ezyhire[18]	Screening tests for intelligence and various aspects of candidates	Provides comprehensive assessment instruments aligned with VirtuHire's evaluation needs	Limited discussion on deep learning capabilities; few field-tested cases.
Codility[19]	Platform for automated coding assessments, including a library of exercises and result analytics	Useful for evaluating technical abilities of VirtuHire candidates	Limited coverage of soft skills and problem-solving abilities.

3.4 Conclusion

This literature review covers three main stages of AI-driven recruitment, which include: first, the application of machine learning and NLP techniques with the use of resume parsing and skill matching; second, the ethical aspect of bias and fairness in the AI system itself; and lastly, the impact of tools related to AI such as chatbots and technical assessments in improving the recruitment process. We explored how AI enhances efficiency in the evaluation of candidates through the concept of automation while papers on bias and fairness lay out the need for ethical considerations when designing AI recruitment systems. Applications discussed give insight into real-world tools that make recruitment smooth, providing a robust base for the development of VirtuHire.

Chapter 4 Software Requirement Specifications

This section provides a comprehensive analysis of the software requirements for our project. It covers a detailed list of features, functional requirements, and quality attributes, as well as non-functional requirements and relevant assumptions. Additionally, it outlines the necessary hardware and software, includes use cases, and provides an overview of the graphical user interface. All essential diagrams and figures are presented, alongside a thorough risk analysis for the project, *VirtuHire*

4.1 List of Features

The following features will be available in the system:

- A simple interface for recruiters to manage, schedule, and update interviews with candidates.
- Allows recruiters to upload, update, and delete resumes in both PDF and DOC formats.
- Provides a platform to specify job positions and manage hiring types (assisted or fully automated).
- Automates the CV screening process using AI, enabling efficient analysis and evaluation against job criteria.
- Offers candidates the ability to log in, set or update interview timings, and track the status of their applications.
- Supports candidates in initiating disputes and requesting Face-to-Face (F2F) interviews if necessary.
- Enables recruiters to send email notifications to applicants, including acceptance, rejection, and interview details.
- Provides a feedback mechanism for both candidates and recruiters to review and assess service quality.
- Generates detailed reports on CV screening and interview results, assisting recruiters in decision-making.
- Allows for the storage and management of interview transcripts, CVs, reports, and feedback.
- Manages payment methods, payment details, history, and user credits for recruiters.
- Provides a dispute management system for monitoring and resolving applicant disputes.
- A Super User interface to manage users, handle finances, and oversee service usage.

4.2 Functional Requirements

The following functional requirements must be met by the system:

1. **User Authentication and Management:**
   - Users (recruiters, candidates, super users) must be able to sign up, log in, and manage their accounts.
   - Role-based access control for recruiters, candidates, and Super Users.
2. **Resume Management:**
   - Recruiters must be able to upload, update, delete, and manage resumes in PDF and DOC formats.
3. **Interview Scheduling and Management:**
   - Recruiters must be able to schedule, update, and delete interviews.
   - Candidates must be able to set, update, or deny interview timings.
4. **CV Screening and Analysis:**
   - The system must automate CV screening using AI models and provide recruiters with reports based on job criteria.
   - Recruiters should be able to initiate and monitor CV screening.
5. **Communication and Notifications:**
   - The system must send email notifications to candidates (accept/reject/F2F interview requests).
   - The system must generate email drafts for candidate communication.
6. **Dispute Management:**
   - Candidates should be able to initiate disputes, and recruiters should be able to manage and resolve disputes.
   - F2F interview requests should be supported for disputed candidates.
7. **Reports and Feedback:**
   - The system should generate detailed reports for CV screening and interview evaluations.
   - Feedback mechanisms must be available for both candidates and recruiters to review ser-

vices.

8. **Payment System:**
   - Recruiters should be able to manage payment details, view payment history, and access user credits.
   - The system must securely handle payment information.
9. **Super User Features:** The Super User must be able to manage user accounts, oversee finances, and monitor system usage.

4.3 Quality Attributes

The system will adhere to the following key quality attributes to ensure optimal performance and user experience:

- **Performance:** The system will deliver fast and efficient processing of CVs and interview scheduling, minimizing latency even under heavy user loads.
- **Scalability:** Designed to scale seamlessly, the system will accommodate growing numbers of users, resumes, and interviews without compromising performance.
- **Security:** The platform will ensure robust data protection through encryption, secure authentication, and adherence to data privacy standards.
- **Reliability:** The system will ensure continuous availability and functionality, minimizing downtime and service interruptions.
- **Usability:** A user-friendly interface will enable recruiters and candidates to easily navigate and use the platform with minimal learning curve.
- **Maintainability:** The system's architecture will allow for easy updates and modifications to meet evolving user and project requirements.
- **Compliance:** The platform will comply with applicable legal standards, such as data protection regulations, ensuring lawful handling of personal information.

4.4 Non-Functional Requirements

The following non-functional requirements must be adhered to:

1. **Performance:**

- The system should process CV screenings within a reasonable time, even under high loads.
- The platform should handle multiple users (recruiters and candidates) simultaneously without lag.

2. **Scalability:** The system must be scalable to accommodate a growing number of users, resumes, and interviews.
3. **Security:**
   - User authentication and data access control should be secure, with sensitive data such as payment information encrypted.
   - The system must ensure data privacy and protection, especially for user credentials, CVs, and reports.
4. **Reliability:**
   - The system must be reliable, ensuring that core features such as CV screening, interview scheduling, and report generation function correctly and consistently.
   - System downtime must be minimal to avoid disruptions in the recruitment process.
5. **Usability:**
   - The interface should be intuitive and easy to use for both recruiters and candidates, requiring minimal training.
   - The system should offer clear and easy-to-understand navigation.
6. **Maintainability:** The system architecture must allow for easy maintenance and updates as the platform evolves.
7. **Compliance:** The system must comply with relevant data protection laws and regulations.
8. **Availability:** The system should have high availability to ensure that recruiters and candidates can access the platform 24/7, except during scheduled maintenance.

4.5 Assumptions

- **Stable Internet Connection:** It is presumed that both recruiters and candidates will have reliable internet connectivity to effectively access the platform, conduct interviews, and submit necessary documents.
- **Accurate Data Entry:** The system operates under the assumption that the information provided

by recruiters and candidates, including CVs, interview details, and feedback, will be accurate and complete.

- **Device Compatibility:** It is expected that users will utilize modern web browsers and devices that fulfill the minimum system requirements to access the platform.
- **Compliance with Data Privacy Regulations:** The platform operates on the assumption that all submitted data will adhere to applicable data protection laws, with users providing the necessary consent for data processing.
- **Reliability of Third-Party Integrations:** It is assumed that any external services or APIs utilized by the system, such as email services, payment gateways, or machine learning libraries, will remain stable and operational.
- **User Authentication Security:** It is presumed that users will be able to log in securely using valid credentials, ensuring that the authentication process effectively prevents unauthorized access.

4.6 Use Cases

This section lists use cases or scenarios from the use-case model if they represent some significant, central functionality of the final system, or if they have a large architectural coverage—they exercise many architectural elements, or if they stress or illustrate a specific, delicate point of the architecture. The table **??** provides a simple use case.

4.6.1 Use Case 1: Sign Up

Name	Sign Up		
Actors	Recruiter,Candidate		
Summary	The user shall provide their Email,username, Password and Full Name on the Sign up form, and after successful verification, redirect the user to the Dashboard.		
Pre-Conditions	The user must not already be logged in. The user must not already be a user.		
Post-Conditions	The user's session is successfully established and shall be redirected to the Dashboard.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user navigates to the Sign up page.	2	The Sign Up form is displayed asking for Full Name, email,username and password.
2	The user enters valid email and password.	4	The system verifies the email/username and password, establishes a session for the user and redirects the user to the dashboard.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Invalid email or password entered.

4.6.2 Use Case 2: Login

Name	Login
Actors	Recruiter, Candidate, Super User
Summary	The user shall enter their Email/Username and Password. After successful verification, the user is redirected to the respective dashboard.
Pre-Conditions	The user must be registered on the platform. The system must be available and the user is not already logged in.
Post-Conditions	The user is successfully logged in and redirected to the dashboard.
Special Requirements	None

Basic Flow			
Actor Action		**System Response**	
1	The user navigates to the Login page.	2	The Login form is displayed asking for Email/Username and Password.
2	The user enters valid Email/Username and Password.	3	The system verifies the credentials, establishes a session for the user and redirects the user to the dashboard.
Alternative Flow			
3	The user enters invalid Email/Username or Password.	4-A	The system responds with an error message: Invalid Email/Username or Password.
4	The user clicks on the ”Forgot Password” link.	4-B	The system prompts the user to enter their registered email.
5	The user enters their registered email and submits.	5-A	The system sends a password reset link to the provided email.
6	The user resets their password using the link received in their email.	6-A	The system confirms the password reset and redirects the user to the login page.

4.6.3 Use Case 3: Payment Method

Name	Payment Method		
Actors	Recruiter		
Summary	The recruiter selects a payment method to process the payment for the services. After successful payment, the system confirms the transaction and provides a receipt.		
Pre-Conditions	The recruiter must be logged in. The recruiter must have selected a service that requires payment.		
Post-Conditions	The payment is successfully processed, and the recruiter receives a confirmation message and receipt.		
Special Requirements	Support for multiple payment methods. Compliance with payment processing standards.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the payment page.	2	The payment options are displayed.
3	The recruiter selects a payment method.	4	The system prompts the recruiter to enter payment details.
5	The recruiter enters payment details and confirms the payment.	6	The system processes the payment with the selected method.
7	The system confirms the payment status and generates a receipt.	8	The recruiter receives a confirmation message and a receipt.
Alternative Flow			
1-A	The recruiter cancels the payment.	2-A	The system navigates back to the service selection page.
5-A	The recruiter enters invalid payment details.	6-A	The system displays an error message: ”Invalid payment details. Please try again.”
7-A	Payment fails due to a processing error.	8-A	The system displays an error message: ”Payment could not be processed. Please check your details or try another payment method.”

4.6.4 Use Case 4: Schedule Interview

Name	Schedule Interview
Actors	Recruiter, AI Bot
Summary	The recruiter specifies a time slot for the interview, allowing the candidate to take the interview at their convenience. The AI bot conducts the interview during the selected time.
Pre-Conditions	The recruiter must be logged in. The system must have available time slots for scheduling the interview.
Post-Conditions	The interview slot is successfully scheduled, and the candidate is notified of the interview time.
Special Requirements	The system should allow the recruiter to specify multiple time slots. The AI bot must be capable of conducting the interview autonomously.

Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the interview scheduling page.	2	The system displays available time slots for scheduling.
3	The recruiter selects a time slot and specifies the candidate's details.	4	The system confirms the selected time slot and candidate details.
5	The recruiter submits the scheduling request.	6	The system schedules the interview and notifies the AI bot to prepare for the session.
7	The system sends a confirmation notification to the candidate.	8	The candidate receives the interview details and instructions.
Alternative Flow			
1-A	The recruiter cancels the scheduling process.	2-A	The system navigates back to the previous page without scheduling.
3-A	The recruiter selects an unavailable time slot.	4-A	The system displays an error message: "Selected time slot is unavailable. Please choose another one."
5-A	The system fails to schedule the interview due to a processing error.	6-A	The system displays an error message: "Unable to schedule the interview. Please try again later."

4.6.5 Use Case 5: Update/Delete Interview Schedule

Name	Update/Delete Interview Schedule
Actors	Recruiter, AI Bot
Summary	The recruiter can update or delete a previously scheduled interview. The AI bot will not conduct the interview if deleted, and updates will notify the candidate.
Pre-Conditions	The recruiter must be logged in. An interview schedule must exist to update or delete.
Post-Conditions	The interview schedule is successfully updated or deleted, and the candidate is notified of the change.
Special Requirements	The system should allow the recruiter to view current schedules. Notifications should be sent to the candidate if updates or deletions occur.

Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the scheduled interviews page.	2	The system displays a list of scheduled interviews.
3	The recruiter selects an interview to update or delete.	4	The system confirms the selection and displays options to update or delete.
5	The recruiter chooses to update the interview schedule.	6	The system prompts the recruiter for new schedule details.
7	The recruiter enters the new schedule details and submits.	8	The system updates the interview schedule and notifies the AI bot and candidate.
9	The recruiter chooses to delete the interview schedule.	10	The system confirms the deletion request and removes the interview from the schedule.
11	The system sends a notification to the candidate about the update or deletion.	12	The candidate receives the notification regarding the changes.
Alternative Flow			
1-A	The recruiter cancels the update/delete process.	2-A	The system navigates back to the scheduled interviews page without changes.
3-A	The recruiter attempts to delete a non-existent interview.	4-A	The system displays an error message: ”Interview schedule not found.”
5-A	The recruiter tries to update an interview with invalid details.	6-A	The system displays an error message: ”Invalid schedule details. Please try again.”

4.6.6 Use Case 6: Specify Job Position

Name	Specify Job Position		
Actors	Recruiter		
Summary	The recruiter specifies a job position in the system by filling out the required fields in the job posting form. After submission, the job position is added to the system for candidates to view and apply.		
Pre-Conditions	The recruiter must be logged in to the system. The recruiter must have the necessary permissions to create job postings.		
Post-Conditions	The job position is successfully added to the system, and candidates are able to view it in the job listings.		
Special Requirements	The job posting form must validate all fields and check for duplicates. Notifications should be sent to the recruiter confirming the job position is posted.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the job posting page.	2	The job posting form is displayed, requesting job title, description, requirements, and other details.
3	The recruiter fills in all required fields and submits the form.	4	The system validates the input and adds the job position to the database if valid.
5	The system sends a confirmation notification to the recruiter.	6	The job position is now visible to candidates in the job listings.
Alternative Flow			
1-A	The recruiter submits the form with invalid or missing information.	2-A	The system responds with an error message indicating which fields need correction.
3-A	The recruiter tries to submit a duplicate job position.	4-A	The system displays an error message: ”Job position already exists.”

4.6.7 Use Case 7: Add/Remove CVs

Name	Add/Remove CVs
Actors	Recruiter
Summary	The recruiter uploads or removes CVs from the system to be screened by the AI bot. Uploaded CVs will be available for the AI to analyze and assess, while removed CVs will no longer be processed.
Pre-Conditions	The recruiter must be logged in to the system. The recruiter must have the necessary permissions to manage CVs.
Post-Conditions	The system updates the list of CVs, either adding or removing the selected CVs, and the AI bot is notified to process the new entries accordingly.
Special Requirements	The system should validate the CV format during upload and provide feedback. The recruiter should receive notifications confirming the upload or removal of CVs.

Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the CV management page.	2	The system displays options to upload or remove CVs.
3	The recruiter selects the option to upload CVs and chooses the files to upload.	4	The system validates the selected files and uploads them, adding them to the CV database.
5	The system confirms the successful upload and notifies the recruiter.	6	The uploaded CVs are now available for the AI bot to screen.
7	The recruiter selects CVs to remove from the system.	8	The system confirms the removal and updates the CV database accordingly.
9	The system notifies the recruiter of the successful removal of the CVs.	10	The removed CVs are no longer available for the AI bot to screen.
Alternative Flow			
1-A	The recruiter tries to upload a CV in an invalid format.	2-A	The system responds with an error message indicating the invalid file format.
3-A	The recruiter tries to remove a CV that does not exist in the system.	4-A	The system displays an error message: ”CV not found in the system.”

4.6.8 Use Case 8: Start CV Screening

Name	Start CV Screening		
Actors	Recruiter, AI Bot		
Summary	The recruiter prompts the AI bot to start the screening process for the uploaded CVs. The system notifies the recruiter once the screening is completed.		
Pre-Conditions	The recruiter must be logged in to the system. CVs must have already been uploaded for screening.		
Post-Conditions	The AI bot screens the CVs and generates a report for the recruiter, including shortlisted candidates.		
Special Requirements	The AI bot must be capable of handling multiple CVs at once. Notifications should be sent to the recruiter when the process is complete.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the CV screening page.	2	The system displays the option to start the screening process.
3	The recruiter clicks the “Start Screening” button to initiate the process.	4	The AI bot starts processing the CVs and displays a notification that screening is in progress.
5	The recruiter waits for the process to complete.	6	The system notifies the recruiter that the screening is complete and provides a report of the shortlisted candidates.
Alternative Flow			
1-A	The recruiter attempts to start screening without uploading CVs.	2-A	The system responds with an error message: “No CVs uploaded for screening.”

4.6.9 Use Case 9: View Application Status

Name	View Application Status		
Actors	Recruiter		
Summary	The recruiter views the current status of applications, including CV screening and interview progress for all candidates.		
Pre-Conditions	The recruiter must be logged in to the system. CVs or interviews must have been uploaded or scheduled.		
Post-Conditions	The recruiter is able to view the status of CVs and interviews, such as whether they have been screened, pending, scheduled, or awaiting feedback.		
Special Requirements	The system must display real-time updates for the status of CVs and interviews.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the application status page.	2	The system displays a list of uploaded CVs and interview statuses.
3	The recruiter selects a job position or candidate to view status details.	4	The system displays the detailed status of CV screening and interviews.
Alternative Flow			
1-A	There are no applications or interviews available.	2-A	The system displays a message: ”No applications found.”

4.6.10 Use Case 10: Respond to Candidate

Name	Respond to Candidate		
Actors	Recruiter		
Summary	The recruiter selects a pre-written response (accept, reject, or request for a face-to-face (F2F) interview), and the system sends the selected email to the candidate.		
Pre-Conditions	The recruiter must be logged in. The candidate must have completed the screening process.		
Post-Conditions	The system sends an email to the candidate based on the recruiter's selection (accept/reject/F2F).		
Special Requirements	The pre-written emails must be customizable and must include the candidate's name and job position.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the candidate response page.	2	The system displays the list of candidates and response options (accept, reject, F2F).
3	The recruiter selects the response (accept, reject, or F2F interview).	4	The system generates the pre-written email based on the selected response.
5	The recruiter confirms the selection and sends the email.	6	The system sends the email to the candidate and updates the candidate's status.
Alternative Flow			
1-A	The recruiter does not select a response or exits the page without confirming.	2-A	The system does not send any email and returns to the previous state.

4.6.11 Use Case 11: View Dispute

Name	View Dispute		
Actors	Recruiter		
Summary	The recruiter views disputes raised by candidates regarding interview outcomes or any issues during the interview process. The system displays the details of the dispute for further action.		
Pre-Conditions	The recruiter must be logged in. The candidate must have raised a dispute.		
Post-Conditions	The recruiter is able to view the dispute details and decide on further actions (resolve, investigate, etc.).		
Special Requirements	The system should allow the recruiter to view all necessary details such as dispute description, interview details, and timestamps.		
Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the disputes management page.	2	The system displays the list of disputes raised by candidates.
3	The recruiter selects a dispute to view the details.	4	The system shows the dispute details such as candidate information, interview details, and dispute reason.
Alternative Flow			
1-A	No disputes have been raised by any candidates.	2-A	The system displays a message indicating that no disputes are available.

4.6.12 Use Case 12: Update/Deny/Set Interview Time

Name	Update/Deny/Set Interview Time		
Actors	Candidate		
Summary	The candidate updates, denies, or sets an interview time based on the interview invitation received.		
Pre-Conditions	The candidate must have received an interview invitation and must be logged in to the system.		
Post-Conditions	The interview is either confirmed within the time slot, or the recruiter is notified of the denial.		
Special Requirements	The system should allow the candidate to select alternate times within the time slot provided.		
Basic Flow			
Actor Action		**System Response**	
1	The candidate logs into the system and navigates to the interview invitations page.	2	The system displays list of interview invitations with their proposed time slots.
3	The candidate selects an invitation and chooses to accept and choose a time slot or deny interview.	4	If accepted, the system confirms the interview and notifies the recruiter. If denied, the recruiter is notified.
Alternative Flow			
3-A	The candidate fails to propose a time within the time slot.	4-A	The system sends a notification to the candidate to fill slot for interview .

4.6.13 Use Case 13: AI Bot Conducts Interview

Name	AI Bot Conducts Interview		
Actors	Candidate, AI Bot		
Summary	The candidate participates in an interview conducted by an AI bot. The AI bot asks a series of predefined or dynamic questions, records the candidate's responses, and analyzes the responses for evaluation.		
Pre-Conditions	The candidate has scheduled the interview and is logged in at the time of the interview. The system should verify that the candidate has the necessary equipment for the interview (e.g., microphone, camera).		
Post-Conditions	The interview is recorded, responses are analyzed, and a report is generated and submitted to the recruiter for further review.		
Special Requirements	The AI bot should ensure the quality of the recording, provide live feedback to the candidate if technical issues arise, and confirm that the interview is completed successfully.		
Basic Flow			
Actor Action		**System Response**	
1	The candidate logs into the system and navigates to the interview section.	2	The system displays the scheduled interview details and provides a link to start the interview.
3	The candidate clicks the start interview link.	4	The AI bot greets the candidate, checks the candidate's equipment, and begins the interview.
5	The AI bot asks a series of questions and records the candidate's responses.	6	The AI bot evaluates the responses based on predefined criteria, stores the recording, and generates an analysis report.
7	The AI bot concludes the interview and thanks the candidate.	8	The system saves interview report and sends it to the recruiter for review.
Alternative Flow			
5-A	The AI bot detects a technical issue (e.g., poor audio or video quality) during the interview.	6-A	The system prompts the candidate to resolve the issue (e.g., checking mic or camera) and retry after fixing.
5-B	The candidate encounters network issues and gets disconnected.	6-B	The system pauses the interview and allows the candidate to reconnect and resume from where they left off.

4.6.14 Use Case 14: View Application Status

Name	View Application Status		
Actors	Candidate		
Summary	The candidate checks the status of their application, which can be either ”Approved”, ”Rejected”, or ”Under Process”.		
Pre-Conditions	The candidate has submitted an application and is logged into the system.		
Post-Conditions	The candidate views their current application status. No other actions are required.		
Special Requirements	The system should provide accurate and real-time status updates.		
Basic Flow			
Actor Action		**System Response**	
1	The candidate logs in and navigates to the ”Application Status” section.	2	The system displays the candidate’s application status (Approved, Rejected, or Under Process).
3	The candidate views their current status.	-	No further system response.
Alternative Flow			
1-A	The system detects no status update yet for the candidate’s application.	2-A	The system displays a message: ”Your application is still under review.”

4.6.15 Use Case 15: Initiate Dispute or Request F2F Interview

Name	Initiate Dispute or Request F2F Interview		
Actors	Candidate		
Summary	The candidate initiates a dispute with the system regarding the interview or application process and provides reasons. Optionally, they can request a face-to-face (F2F) interview.		
Pre-Conditions	The candidate must be logged into the system and have undergone an interview or application process.		
Post-Conditions	The dispute is successfully recorded, and the system sends a notification to the recruiter for review. If a request for F2F is made, the recruiter will review and respond.		
Special Requirements	The system should capture and store the dispute reasons and any requested action (e.g., F2F interview).		
Basic Flow			
Actor Action		**System Response**	
1	The candidate navigates to the “Initiate Dispute” or “Request F2F” page.	2	The system displays a form to capture the reason for the dispute or F2F request.
3	The candidate provides reasons for the dispute, and optionally requests a face-to-face interview.	4	The system saves the dispute and sends a notification to the recruiter for review.
5	The candidate submits the form.	6	The system confirms that the dispute/request has been submitted successfully.
Alternative Flow			
1-A	The candidate tries to initiate a dispute without having undergone an interview or application process.	2-A	The system displays an error message: ”You cannot initiate a dispute at this stage.”
3-A	The candidate does not provide sufficient reason for the dispute or request.	4-A	The system prompts the candidate to provide more information before submitting.

4.6.16 Use Case 16: Manage Subscription Plan

Name	Manage Subscription Plan
Actors	Recruiter
Summary	The recruiter manages their subscription plan by viewing their current plan details, upgrading to a higher plan, or canceling the existing plan.
Pre-Conditions	The recruiter must be logged in. The recruiter must have an active subscription plan.
Post-Conditions	The recruiter's plan is updated, canceled, or remains unchanged based on the action taken. If canceled, access to services is revoked after the current billing period.
Special Requirements	The system should ensure that any plan upgrade or cancellation follows proper payment and notification protocols.

Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the ”Manage Subscription Plan” page.	2	The system displays the current subscription details, including plan type, cost, and next billing date.
3	The recruiter selects ”View Plan”, ”Upgrade Plan”, or ”Cancel Plan”.	4	The system displays the chosen action's form or confirmation messages.
5	If upgrading, the recruiter chooses a higher plan and confirms the payment.	6	The system processes the upgrade and sends a confirmation email with updated plan details.
7	If canceling,the recruiter confirms the cancellation.	8	The system confirms cancellation and sends an email notification, informing that services will be available until the end of the billing cycle.
Alternative Flow			
1-A	The recruiter attempts to upgrade to a plan without sufficient funds or valid payment method.	2-A	System displays an error message prompting recruiter to update payment details.
3-A	The recruiter tries to cancel but has an ongoing service requirement (e.g., active hiring).	4-A	The system prompts the recruiter to complete any ongoing processes before canceling.

4.6.17 Use Case 17: Set Interview Time Slots

Name	Set Interview Time Slots
Actors	Recruiter
Summary	The recruiter sets a time frame (in days or weeks) for candidates to register for interview time slots. The recruiter also activates the registration process during this period.
Pre-Conditions	The recruiter must be logged in. There should be an active hiring process.
Post-Conditions	Candidates can register for available time slots within the defined time frame. The system updates the recruiter with registration status.
Special Requirements	The system should ensure time slot availability is updated in real time as candidates register.

Basic Flow			
Actor Action		**System Response**	
1	The recruiter navigates to the ”Set Interview Time Slots” page.	2	The system displays available time frame options (e.g., days, weeks).
3	The recruiter selects a start date, end date, and daily time slots for interviews.	4	The system confirms the time frame and slot availability, then activates the registration.
5	The recruiter reviews or modifies the time frame or slots if needed.	6	The system updates the schedule accordingly and confirms the changes.
7	The recruiter saves and activates the process.	8	The system activates interview slot registration for candidates and notifies the recruiter.
Alternative Flow			
1-A	The recruiter attempts to set a time frame that overlaps with an existing schedule.	2-A	System displays an error message and suggests alternative slots.
3-A	The recruiter attempts to set slots outside of business hours.	4-A	System prompts the recruiter to choose slots within allowable hours.

4.7 Hardware and Software Requirements

This section outlines the necessary hardware and software components required for the successful development and deployment of the project.

4.7.1 Hardware Requirements

We will require a computer system with the following capabilities for the developement of the application.

- The system must be powerful enough to run software such as React, Node.js, MongoDB, and Django, and should be able to handle the execution of AI models efficiently.
- It should be able to run both client-side and server-side of the application seamlessly.
- A fast and reliable internet connection is essential for development and testing.
- The system should also be equipped with a functional microphone and camera for testing purposes.

4.7.2 Software Requirements

Development Environment:

- Node.js
- MongoDB
- React.js
- Next.js
- Express.js
- Django
- Tailwind CSS

Integrated Development Environment (IDE):

- Visual Studio Code or any other preferred code editor
- MongoDB Compass for database management

Programming Languages:

- Javascript

- Python
- HTML
- CSS

Version Control:

- Git and GitHub for source code management

Testing Tools:

- Postman for API testing
- Jest or Mocha for unit testing

Deployment:

- Docker for containerization
- Cloud service provider (e.g., AWS, Firebase, Vercel) for hosting the application

Additional Libraries/Frameworks:

- Natural Language Processing libraries (e.g., spaCy, NLTK)
- Machine Learning libraries (e.g., TensorFlow, PyTorch)
- Authentication libraries (e.g., Passport.js or Firebase)
- Meta's LLaMA (Large Language Model)

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, with reference to the users. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers.

4.9 Database Design (if required)

4.9.1 ER Diagram

4.9.2 Data Dictionary

Table 4.1: Data Dictionary

Data Element	Description	Type	Format/Constraints
Candidate ID	A unique identifier assigned to each candidate within the system.	String	Auto-generated, unique.
Candidate Name	The full name of the candidate as provided in their CV.	String	Max length: 100 characters.
Candidate Email	The email address extracted from the candidate's CV, used for communication.	String	Must follow email format (e.g., example@domain.com).
CV	The candidate's CV document uploaded by the recruiter, containing the candidate's professional and academic details.	File (PDF)	Max size: 5MB. Accepted formats: PDF, DOCX.
CV Rating	A rating score given by the AI system after analyzing the candidate's CV for job fit.	Integer	Range: 0-100 (percentage).
Interview Time Slot	Time slots created by the recruiter for candidates to schedule their interview. Each slot includes date and time range available for the candidate within a specified time frame.	Date/Time	Format: `YYYY-MM-DD HH:MM:SS`. Slots are within recruiter-set constraints (e.g., 7 days).
Interview Invite	The email invite sent to the candidate with a link to schedule their interview.	String	Auto-generated, sent to candidate's email.
Interview Rating	A rating score assigned by the AI system after conducting and analyzing the candidate's interview performance.	Integer	Range: 0-100 (percentage).
AI Interview Results	The analysis results produced by the AI bot after the candidate's interview.	Text	Includes key feedback and recommendations.
Acceptance Status	Indicates whether the candidate was accepted or rejected after the screening process.	Enum	Values: "Accepted", "Rejected".
Screening Dispute	A request by the candidate to dispute the AI screening or interview results.	Boolean	True/False. Can be raised within 7 days of receiving results.
Face-to-Face Request	A request submitted by the candidate for a face-to-face interview instead of an AI interview.	Boolean	True/False. Can be requested before scheduling the AI interview.
Recruiter ID	A unique identifier for the recruiter responsible for managing CVs and time slots.	String	Auto-generated, unique.
Recruiter Name	Full name of the recruiter responsible for managing candidate CVs and interview scheduling.	String	Max length: 100 characters.
CV Upload	The action performed by the recruiter to upload a candidate's CV into the system.	Action	Logged with timestamp.
CV Update	The action performed by the recruiter to update an existing candidate's CV in the system.	Action	Logged with timestamp.
CV Delete	The action performed by the recruiter to delete a candidate's CV from the system.	Action	Logged with timestamp.
Screening Email	Email sent to the recruiter with the AI screening and interview results for each candidate.	String	Auto-generated.
Interview Deadline	The deadline by which the candidate must schedule their interview (e.g., within 7 days of receiving the invite).	Date	Format: `YYYY-MM-DD`.
Recruiter Selection	The final decision made by the recruiter after reviewing the AI screening and interview results.	Enum	Values: "Selected", "Not Selected".

4.10 Risk Analysis

This section highlights the key risks associated with the development, deployment, and ongoing maintenance of the project.

1. Technical Risks

- **Integration Complexities**: Integrating diverse components, such as Node.js, Django, and third-party APIs, may lead to technical challenges, impacting system functionality.
- **Data Security Threats**: The platform handles sensitive user data, making it vulnerable to security breaches if robust security protocols are not maintained.
- **Scalability and Performance**: As user demand grows, the system may encounter performance issues, such as slow response times or server overloads, if not designed with scalability in mind.

2. Business Risks

- **Market Reception**: The platform may struggle to gain market traction due to competition or limited interest from the target audience.

3. Operational Risks

- **Project Mismanagement**: Inadequate communication or project coordination among team members could lead to delays, misaligned expectations, or scope creep.
- **Third-Party Dependencies**: Reliance on external services like payment gateways or email providers may introduce risks if these services face disruptions or policy changes.

4. User Adoption Risks

- **Poor User Experience**: If the platform lacks an intuitive interface, users may face difficulties in navigation, leading to reduced adoption and engagement.
- **Training and Support Needs**: A lack of adequate training or user support could limit the effective use of the platform, resulting in lower user satisfaction.

5. Compliance Risks

- **Regulatory Challenges**: Changes in data privacy regulations may result in non-compliance issues, leading to legal complications. Continuous updates to ensure adherence to regulations are necessary for platform success.

6. Market Risks

- **Competitive Pressure**: The recruitment technology sector is highly competitive, and failure to stay ahead of market trends or enhance features may lead to a loss of competitive advantage.

Chapter 5 Proposed Approach and Methodology

5.1 Introduction

This chapter outlines the proposed approach and methodology for our project, focusing the backend algorithms and processes that power the application's functionality. We aim to integrate various models and tools to streamline cumbersome and resource consuming tasks such as CV screening, shortlisting, scheduling interviews, conducting personality assessments and technical evaluations, building a conversational AI, which will provide a more efficient and smooth recruitment experience. The integration of LLaMA 3 adds significant value, enabling the platform to analyze CVs and responses effectively and generate insightful recommendations for recruiters.

5.2 CV Screening Models

The platform aim to utilize advanced machine learning models to evaluate CVs effectively. These models extract essential information from candidates' resumes and rank them according to specific job criteria, filtering out those who don't meet the appropriate qualifications. With the help of LLaMA 3, the system can generate clear summaries and recommendations for each candidate. This feature empowers recruiters to access a candidate's suitability, making the decision-making process more informed and efficient.

5.3 Personality Assessment (Myers-Briggs)

We aim to integrate the Myers-Briggs personality assessment to gauge candidates' behavioral aspects and compatibility with various roles. The system evaluates the Myers-Briggs results and uses this data alongside technical assessments to construct a holistic well-rounde candidate profile, which provides deeper insights into how an individual might fit into team dynamics and organizational culture.

5.4 Speech-to-Text and AI Bot Interaction

The platform is set to feature speech-to-text technology to transcribe voice interviews, converting audio responses into text data for easier analysis. Additionally, an AI-powered conversational bot, supported by NLP models such as LLaMA 3, simulates human-like interaction during interviews. This bot processes and responds to candidates' text-based answers, offering a smooth and realistic interview experience.

5.5 Use of LLaMA 3 for Analysis and Recommendations

LLaMA 3 plays a vital role in the evaluation process by carefully analyzing candidates' resumes and understanding their context. It provides meaningful and insightful descriptions and relevant recommendations that help recruiters quickly identify standout candidates. By summarizing each candidate's strengths, qualifications, and experiences, LLaMA 3 makes it easier for recruiters to see who might be the best fit for the role. Furthermore, it generates numerical and statistical scores, giving recruiters valuable insights and metrics to enhance their decision-making and prioritize candidates effectively.

5.6 Conclusion

In this chapter, we presented our proposed approach and methodologies aimed at enhancing the recruitment process. By using CV screening models, personality assessments, speech-to-text technology, and the power of LLaMA 3, we are committed to creating a more efficient hiring process. Our integrated AI-driven tools not only automate technical assessments but also provide with engaging, human-like interactions. The application is designed with scalability in mind, our platform is capable to evolve alongside future developments and improvements, making the hiring process smoother and more effective for both recruiters and candidates.

Chapter 6 High-Level and Low-Level Design

This chapter contains the high- and low-level design of the system.

6.1 System Overview

This section provides with the basic overview of the system.

6.1.1 Platform Description

The VirtuHire AI recruitment platform streamlines and optimizes the hiring process by using automation and advanced technology. Recruiters and candidates receive an interactive user interface in this feature that optimizes not only the recruiter's user experience but also the candidate's experience. The resume filtering and analysis module automatically screens resumes according to the recruiters' specified criteria, ensuring only the most relevant candidates are selected. Additionally, for candidates who proceed to interviews, the recruiters receive detailed feedback and analysis, providing them with valuable insights into their performance and suitability.

6.1.2 Interactive User Interface for Recruiters and Candidates

VirtuHire features a responsive, intuitive interface designed to serve both recruiters and candidates. For recruiters, the interface provides real-time data on candidate profiles, filtered resumes, and interview feedback. Candidates benefit from a streamlined application proces, clear communication, and AI-driven assessments that simulate real interview scenarios. The interface ensures seamless interaction between both parties, making the recruitment experience smooth and efficient.

6.1.3 Resume Filtering and Analysis

This strength for this application lies in the depth of resume filtering and analysis that it performs. The platform uses NLP and ML to analyze a resume, rank candidates by pertinent job-specific criteria, and deliver better insights, with LLaMA 3 it can also generate summaries and recommendations to help recruiters quickly see the qualified candidates.

6.1.4 Automated or Semi-Assisted Interviews

VirtuHire offers a completely dynamic interviewing process, and full options for both fully automated and semi-assisted interviews. Candidates will interact directly with an AI-powered conversational bot in fully automated interviews, simulating real interview scenarios. For a more interactive approach, semi-

assisted interviews enable recruiters to conduct live interviews with the AI running in the background. During these sessions, the AI acts as a companion, offering real-time feedback on candidate responses, recommending follow-up questions, and highlighting key areas of interest. This combination of human oversight and AI support allows for a more personalized and efficient interview process, enhancing decision-making for recruiters.

6.1.5 Interview Feedback and Analysis

VirtuHire's interview feedback and analysis feature provides recruiters with a holistic view of each candidate's performance. The AI-driven analysis is not limited to content-related assessments; it incorporates tone, communication style, and technical knowledge as well into the review. This feature will be particularly powerful in identifying key personality traits, cultural fit, and technical capabilities. This therefore enables the recruiters to base their judgment on facts by making data-driven decisions about the candidate's areas of strengths and weaknesses. The system will flag whether a candidate's personality would mesh well with the team dynamics and organizational culture or otherwise potentially mismatch. This way that the system enhances recruitment by giving better insights into a candidate beyond mere technical skills.

6.1.6 Performance and Responsiveness

VirtuHire is designed to be fast and highly responsive, ensuring smooth performance across all its modules. By utilizing the MERN stack for providing a canvas for the application and Django for AI functionalities, the platform can efficiently handle large volumes of data and complex processing tasks without compromising speed. It makes full use of modern technology in terms of MongoDB's database management and Next.js for server-side rendered front-end development to speed up loads and retain real-time updates. The blending with Django enhances the AI capabilities and operations on the platform for efficient handling complex tasks. Moreover, the system is highly scalable, so it can easily bear an increasing number of users without any degradation in performance.

6.2 Design Considerations

This section describes many issues that need to be addressed or resolved before attempting to devise a complete design solution.

6.2.1 Assumptions and Dependencies

The design of VirtuHire is based on the following assumptions and dependencies.

6.2.1.1 Related software or hardware:

The platform assumes that users will have access to modern browsers and internet-enabled devices capable of handling high-end web applications. It is designed to run on servers optimized for high-load processing, as it involves AI and ML tasks. Additionally, VirtuHire is fully responsive and works fine on all screen sizes: from a big desktop screen to a tablet or small smartphone screen, the platform is able to produce a seamless user experience. It holds excellent performance and responsiveness on mobile devices in which users can seamlessly navigate the application regardless of the used screen size.

6.2.1.2 Operating systems:

VirtuHire is cross-platform compatible with Windows, macOS, and Linux, and responds with a mobile and tablet version, where the screen is small. The system maintains great consistency in performance on any Android or iOS operating system and any screen size.

6.2.1.3 End-user characteristics:

The system presumes recruiters and candidates will have at least basic technical skills to tap their way through the system. It further presumes recruiters will understand how to interpret the feedback from the system.

6.2.1.4 Possible/probable changes in functionality:

Future versions or updates may introduce changes in functionalities as new AI technologies emerge. The system design is modular with the future scalability or feature upgrade that can be done without significant overhauls.

6.2.2 General Constraints

This section outlines the significant limitations and constraints impacting the design of VirtuHire's software. These constraints are crucial for ensuring the system operates effectively and meets user expectations:

6.2.2.1 Hardware or Software Environment:

The platform must function efficiently across various hardware configurations, including low-spec devices, as well as older browsers that may not support modern web technologies. This requires optimizing resource usage to maintain performance, particularly on older or less powerful machines. Additionally, VirtuHire must ensure responsiveness across different screen sizes, from large desktop monitors

to smaller tablets and smartphones.The design should adapt to devices having varied resolutions and aspect ratios, and users will never know if a certain device is being used because the application will intelligently adjust to whatever the user is using. This focus on adaptability ensures that all users can effectively access and interact with the platform, regardless of their hardware or browser capabilities..

6.2.2.2 End-User Environment:

Users may work in different environments with varying internet connection speeds. The application should be robust enough to handle bandwidth conditions of low bandwidth and still deliver the core functionalities.

6.2.2.3 Availability or Volatility of Resources:

The AI and ML models the system relies on also need a steady level of access to quality training data. Variability in data availability might impact the model's performance and stability.

6.2.2.4 Standards Compliance:

VirtuHire should comply with relevant industry standards and laws on data privacy such as GDPR. Compliance impacts the design since it must be capable of practicing safe data handling practices.

6.2.2.5 Interoperability Requirements:

The platform must integrate seamlessly with various third-party services and APIs. Ensuring compatibility can complicate the design and increase development time.

6.2.2.6 Interface/Protocol Requirements:

Standardized communication protocols for exchanging data between applications must be strictly adhered to. Incompatibilities could hinder interoperability with external systems, potentially reducing functionality and complicating future maintenance activities. Additionally, maintaining standard UI features and workflows is essential to ensure a consistent user experience. Adhering to familiar design patterns and interface guidelines will help users navigate the platform efficiently, thereby enhancing usability and adoption rates.

6.2.2.7 Data Repository and Distribution Requirements:

The system needs to handle large volumes of data effectively. The choice of database technology impacts how data is stored, retrieved, and distributed. It modifies the performance and scalability of the system depending on how the data will be held, retrieved, and shared. The flexible schema design of

MongoDB makes it easy to manage dynamic structures of data and user growth very effectively, thus easy to implement or set up the system using such a database technology.

6.2.2.8 Security Requirements:

The platform handles sensitive data related to both candidates and the organization, which necessitates robust security measures to prevent potential data breaches. This includes implementing secure authentication processes, such as multi-factor authentication, to verify user identities effectively. Additionally, data storage must be encrypted to protect sensitive information from unauthorized access. These security requirements introduce further design considerations and performance constraints on the system, ensuring that security measures do not compromise the overall user experience or system efficiency.

6.2.2.9 Performance Requirements:

The system is expected to provide quick response times and handle multiple users simultaneously. This requires careful design and optimization to ensure optimal performance.

6.2.2.10 Network Communications:

VirtuHire heavily relies on stable internet connections for optimal performance. Disruption in network quality will impact the experience of the user and the speed at which data is exchanged with the server.

6.2.2.11 Verification and Validation Requirements:

The design must incorporate thorough testing strategies to guarantee the software's reliability and functionality. This includes implementing crash analytics and system feedback mechanisms that provide insights into performance issues and user experience. Such processes may impact development timelines and resource allocation, as they require careful attention to ensure that both functionality and maintenance needs are adequately addressed.

6.2.3 Goals and Guidelines

In designing VirtuHire, several goals and guidelines have been established to ensure the system is effective, user-friendly, and scalable. These guiding principles shape the overall development approach and are critical to the project's success.

6.2.3.1 KISS Principle

The KISS principle ("Keep It Simple, Stupid!") emphasizes simplicity in design and functionality. By focusing on straightforward features and intuitive interfaces, the system becomes more accessible to

users of varying technical expertise. This approach minimizes the learning curve and encourages user adoption, ensuring that both recruiters and candidates can efficiently navigate the platform.

6.2.3.2 Performance Optimization

A strong emphasis is placed on performance, balancing speed and memory usage. The goal is to create a responsive application that quickly processes data and provides real-time feedback, which is crucial for user satisfaction. Optimizing performance also ensures that the system can handle a large volume of concurrent users, especially during peak usage times.

6.2.3.3 Consistency with Existing Products

The design aims to create a user experience that feels familiar and intuitive, resembling existing recruitment platforms. This familiarity helps reduce resistance to adoption, as users are more likely to engage with a system that aligns with their previous experiences. By implementing standard UI features and workflows, VirtuHire can enhance usability and streamline the recruitment process.

6.2.3.4 Scalability and Flexibility

The system is designed with scalability in mind, allowing it to grow and adapt alongside the changing needs of organizations and advancements in AI technologies. This flexibility ensures that VirtuHire can incorporate new features and functionalities without significant overhauls, thus future-proofing the platform and maximizing return on investment.

6.2.3.5 User Engagement and Retention

The crucial aim of VirtuHire is to have high user engagement and retention while delivering people customized experiences and valuable insights. Features that track user activity, offer recommendations, and deliver feedback can add a sense of involvement, which might encourage people to go back more frequently to the platform. Furthermore, the incorporation of gamification elements, such as achievement badges for the completion of profiles or successful hires, encourages users to make more proactive approaches and spend more time on the platform.

6.2.3.6 Persuasive Design Elements

The design includes persuasive elements aimed at encouraging users to invest in premium features or services, by emphasizing the value and advantages of these features through strategic positioning, engaging visuals, and clear communication, the platform aims to boost conversion rates. Additionally,

showcasing social proof, such as testimonials and case studies from happy users, can further encourage potential customers to invest in VirtuHire's offerings.

6.2.4 Development Methods

Our development approach for VirtuHire follows the Scrum framework, allowing us to efficiently manage tasks and adapt to changing requirements. As a team of three, each member brings a specific strength in one area while having a foundational understanding of other skills. We share the roles of Scrum Master and Product Owner among ourselves, ensuring that everyone is involved in facilitating the Scrum process and prioritizing project goals.

We divide modules and components based on our expertise and provide assistance to one another when challenges arise. This collaborative environment fosters a flexible workflow, enabling us to incorporate feedback and make necessary adjustments throughout the development process while adhering to the fixed major requirements of the project. Scrum's iterative nature supports our ongoing progress and helps us maintain clear communication and accountability among team members.

6.3 System Architecture

The architecture of Virture Hire is designed as a modular system, broken down into distinct layers that separate the front-end interface from business logic, AI capabilities, and data persistence. This layered architecture allows for a scalable, maintainable, and efficient platform for recruitment processes, leveraging modern web technologies, AI, and databases.

At a high level, the system is partitioned into the following layers:

Front-end Layer (Node.js):

The front-end is responsible for handling user interactions (candidates and recruiters). It provides user authentication, dashboard management, interview system interface, data manipulation, and report management functionalities. These elements work together to present and manage interactions efficiently, interfacing with both candidates and recruiters.

Responsibilities:

- Manage user sessions and access control.
- Display dashboards and manipulate available recruitment data.
- Handle interview system-related interactions, such as scheduling or evaluation feedback.
- Generate reports for recruiters.

Business Logic Layer (Django):

The business logic is housed in the middle layer, managing core systems like CV management, email management, and interview systems. This layer integrates with AI agents for natural language processing (NLP), enabling enhanced CV screening, interview analysis, and data manipulation tasks. The Django framework manages this layer, focusing on the orchestration of business processes.

Responsibilities:

- Handle CV and interview management, with AI integration for advanced functionalities.
- Manage communication between subsystems such as speech and text manipulation, and email systems.
- Act as a mediator between the front-end interface and the underlying AI and data persistence layers.

LLM Integration Layer:

This layer is responsible for handling complex AI functionalities, particularly related to natural language processing (NLP). It works in conjunction with the business logic layer, where the AI models provide CV screening and interview analysis. This setup supports the automation of candidate evaluation through AI-driven data processing.

Responsibilities:

- Process and analyze candidate interviews.
- Screen CVs through NLP manipulation.

Data Persistence Layer (MongoDB):

This layer is responsible for storing and managing the system's data, such as CVs, interview results, and user data. MongoDB is used to handle non-relational data storage in this component. The data is stored, retrieved, and managed through the data management system integrated into the business logic layer.

Responsibilities:

- Maintain and manage the database, ensuring data integrity.
- Handle interactions between the business logic layer and the persistent storage system.

Rationale for Decomposition:

The system is decomposed into these four primary layers to ensure a separation of concerns and maintain flexibility in development and scaling:

- Node.js is chosen for the front-end due to its ability to handle asynchronous events, making it ideal for real-time user interactions (e.g., interviews).
- Django was selected for the business logic layer due to its strong support for complex web applications and ease of integration with AI systems.
- The LLM Integration Layer handles AI tasks, providing modularity for implementing large language models (LLM) as the system grows.
- MongoDB serves as the data persistence layer due to its flexibility in handling unstructured and semi-structured data like CVs and reports.

This decomposition ensures that each layer handles its responsibilities independently, making the system easier to maintain, debug, and scale. Additionally, this architecture supports modular development, where each layer can be enhanced or modified without affecting the other parts of the system significantly.

6.3.1 System Architecture Design:

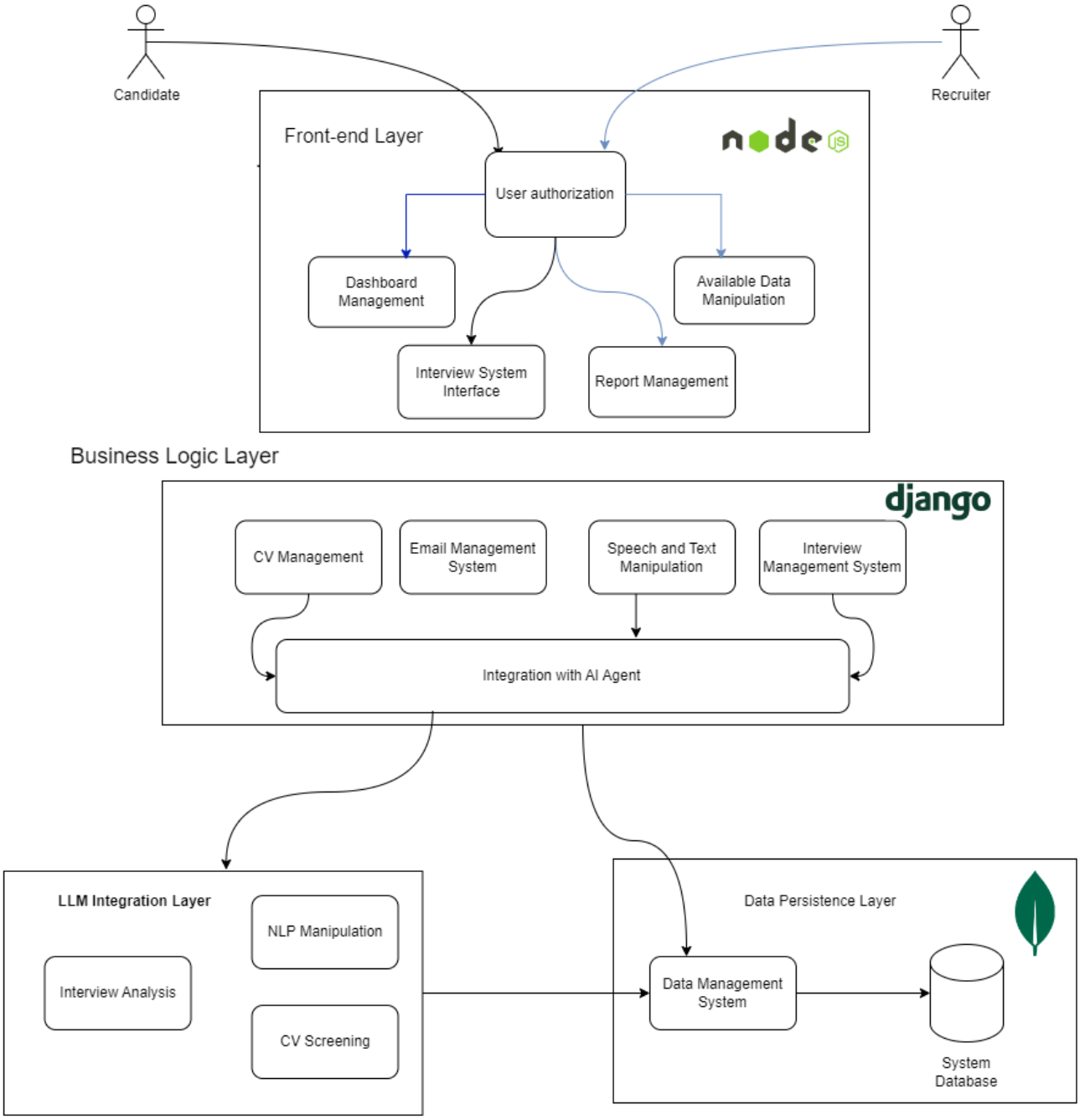

Figure 6.1: System Architecture of VirtuHire Project

6.3.2 Subsystem Architecture

A major part in the Subsystem Architecture is **the Interview Management System** in the Business Logic Layer as it focuses on the following and provides these features.

Real-time Communication:

The system facilitates live communication between the candidate and an AI-powered bot, which conducts the interview. This requires a sophisticated, low-latency, and highly responsive system capable of handling both verbal and textual interactions.

AI Agent Integration:
This system is integrated with an AI agent that can process inputs (such as answers from the candidate) in real-time using natural language processing (NLP) capabilities. The AI not only asks questions but also interprets the candidate's responses for deeper analysis, driving key functionalities like assessment, personality analysis, and interview scoring.

Speech and Text Manipulation:
A major part of the Interview Management System is its ability to manage and manipulate both spoken and written inputs. This could involve speech-to-text conversion for real-time analysis, as well as generating responses or follow-up questions based on the interview's progression.

Business Integration:
The Interview Management System is seamlessly integrated with other business logic components, such as CV Management, where candidate information is pre-loaded, and Email Management, which might trigger follow-up communications or interview summary reports.

Backend AI and NLP Support:
In the LLM Integration Layer, the Interview Analysis and NLP Manipulation components are also directly linked to the Interview Management System, further enhancing the AI's ability to process interview data (both structured and unstructured) for better candidate evaluation.

6.4 Architectural Strategies

Explain all strategies that you use for designing of the Architecture. Describe any design decisions and/or strategies that affect the overall organization of the system and its higher-level structures. These strategies should provide insight into the key abstractions and mechanisms used in the system architecture. Describe the reasoning employed for each decision and/or strategy (possibly referring to previously stated design goals and principles) and how any design goals or priorities were balanced or traded-off. Such decisions might concern (but are not limited to) things like the following:

- Use of a particular type of product (programming language, database, library, etc. ...)
- Reuse of existing software components to implement various parts/features of the system
- Future plans for extending or enhancing the software

- User interface paradigms (or system input and output models)
- Hardware and/or software interface paradigms
- Error detection and recovery
- Memory management policies
- External databases and/or data storage management and persistence
- Distributed data or control over a network
- Generalized approaches to control
- Concurrency and synchronization
- Communication mechanisms
- Management of other resources

6.4.1 Architectural Strategies Strategy-1 name or description

6.4.2 Architectural Strategies Strategy-2 name or description

Each significant strategy employed should probably be discussed in its subsection or (if it is complex enough) in a separate design document (with an appropriate reference here, of course). Make sure that when describing a design decision you also discuss any other significant alternatives that were considered and your reasons for rejecting them (as well as your reasons for accepting the alternative you finally chose). Sometimes it will be most effective to employ the ”pattern format” for describing a strategy.

6.5 Domain Model Diagram

The following Domain Model Diagram shows the Structural Design of VirtuHire.

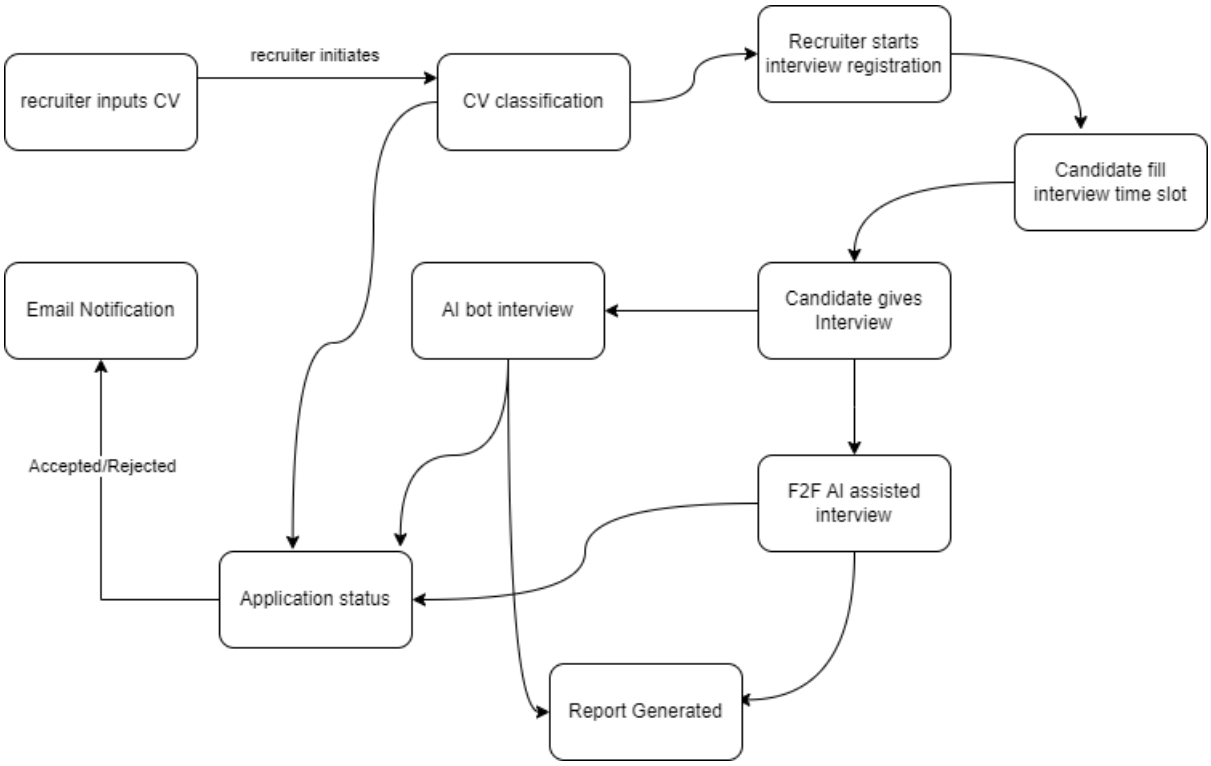

Figure 6.2: System Architecture of VirtuHire Project

6.6 Policies and Tactics

This section outlines the design policies and tactics implemented within the VirtuHire system that impact the finer details of the interface and implementation. These decisions, while not affecting the overall system architecture, play a crucial role in ensuring that the system operates efficiently, is maintainable, and meets project requirements.

6.6.1 Policy/Tactic 1:

Choice of Technology Stack VirtuHire uses a modern technology stack that ensures scalability, ease of development, and flexibility:

Frontend: React/Next.js with Tailwind CSS for building responsive and user-friendly interfaces. Backend: Node.js for database management, and Django for handling AI-related operations. Database: MongoDB was chosen for its scalability and flexibility in handling unstructured data. Alternatives Considered: SQL-based databases were considered, but MongoDB's document model aligns better with the

dynamic nature of recruitment data (e.g., resumes, interview logs, etc.).

6.6.2 Policy/Tactic 2:

Engineering Trade-offs Performance vs. Security: While high performance is essential, special focus is placed on secure authentication (e.g., via Firebase or Passport.js) and secure data storage, especially considering the sensitivity of candidate and recruiter data. Alternatives Considered: Lightweight encryption algorithms were considered but rejected in favor of more robust security methods like AES for enhanced security even if it marginally impacts performance.

6.6.3 Policy/Tactic 3:

Coding Guidelines and Conventions The codebase follows a standardized set of conventions for consistency and maintainability:

Frontend: Follows industry standards for React and Next.js, using the ESLint tool for consistent styling. Backend: Follows PEP 8 (Python) guidelines for Django and ESLint/Prettier for Node.js code.

6.6.4 Policy/Tactic 4:

Testing and Debugging Plans Automated Testing: The platform integrates unit testing using frameworks like Jest (for frontend) and PyTest (for backend Django). Automated tests are run for each new feature before merging code into the main branch. Crash Analytics: Tools like Sentry are used for error tracking and reporting bugs, ensuring that any issues encountered in production are quickly identified and resolved.

6.6.5 Policy/Tactic 5:

Maintainability and Scalability To ensure long-term maintainability:

Version Control: Git is used for version control, with feature branching and regular code reviews to ensure a clean and manageable codebase. Modular Structure: The system's source code is organized in a modular fashion, with each subsystem (e.g., AI processing, user authentication) encapsulated within its directory.

6.6.6 Policy/Tactic 5:

Interface Standards UI Consistency: The presentation layer follows standardized design patterns to ensure a seamless user experience. Tailwind CSS provides reusable components, ensuring a uniform look and feel across the application.

6.6.7 Policy/Tactic 6:

Alternative Approaches Considered Monolithic vs. Microservices Architecture: The project considered adopting a microservices architecture for the backend. However, the team opted for a monolithic structure, given the project's scope, as it simplified development and avoided the added complexity of managing multiple services. Microservices may be reconsidered for future scalability needs. This collection of policies and tactics ensures that VirtuHire remains both performant and secure while adhering to best practices in software design.

Chapter 7 Implementation and Test Cases

For each chapter, provide a paragraph of introduction and at the end, a paragraph of conclusions. (Not the programming code but the algorithmic and procedural details especially related to the hidden/ backend algorithms that are not covered in the design)

7.1 Implementation

Whatever implementation that you have done so far, please elaborate here. Give clear details of the algorithms that were implemented along with the platform and the APIs which were used. NOTE: For FYP-1, this chapter can be changed to description of prototype developed.

7.1.1 Implementation of First Component/Algorithm

Write the implementation of the first component of your system here.

7.2 Test case Design and description

This section will be added in FYP-II. Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail. The Sample test case is provided in table 7.1.

Table 7.1: Sample Test case No.1

<Software Component Name>			
<Reference>			
Test Case ID:	*Reference Number*	**QA Test Engineer:**	*Name of personnel*
Test case Version:	*Version number*	**Reviewed By:**	*Testing Team lead*
Test Date:	*Date*	**Use Case Reference(s):**	*Relation to use cases*
Revision History:	*Refer to previous test case identity (if any)*		
Objective:	*Need and scope of the testing*		
Product/Ver/ Module:	*Refer to overall system being built and the place of this test case in it.*		
Environment:	*Necessary and desired properties of the test environment.*		
Assumptions:	*Assumptions that might affect the testing process.*		
Pre-Requisite:	*Necessary condition that needs to be fulfilled before the test case.*		
Step No.	**Execution description**	**Procedure result**	
	Events being tested.	*Mention software response.*	
Comments:			
Passed Failed Not Executed			

7.3 Test Metrics

Summarize here the common ground of attributes of test case metrics as explained in Table 7.2.

Table 7.2: Sample Test case Matric.No.1

Metric	Purpose
Number of Test Cases	Total number of test cases that you have developed for your system.
Number of Test Cases Passed	The number of test cases that successfully passed
Number of Test Cases Failed	The number of test cases that failed
Test Case Defect Density	(No of test cases failed * 100) No of test cases executed
Test Case Effectiveness	No of defects detected using test cases *100 Total number of defects detected
Traceability Matrix	Traceability is the ability to determine that each the feature has a source in requirements and each requirement has a corresponding implemented feature.

7.3.1 Sample Test case Metric.No.2

Similar to the above table, add here for test metrics 2,3,4.

Chapter 8 User Manual

This chapter will be added in FYP-II. Design a complete user manual of your application and add that in this chapter. NOTE: This Chapter is optional for Research projects but compulsory for Development projects.

Chapter 9 Experimental Results and Discussion

This chapter will be added in FYP-II. Give proper analysis and discussion of experimental results (in plain English text) along with tables of results. For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.

NOTE: This Chapter is optional for Development projects but compulsory for Research projects.

Chapter 10 Conclusions

10.1 aa

10.1.1 dfs

10.1.1.1 dsd

This chapter is mandatory. Give conclusions and summary of the work done. What were your findings and what were the results? Discuss in detail whether the scope of your project was entirely covered or not and whether the objectives of the project were met or not. What challenges did you face and what has been left out and why.

Sum up all the conclusions of all the chapters here to make a final conclusion chapter. Do not repeat any text, just summarize it in different words. Give recommendations for future work also.

For FYP-1 it is mandatory to list down a plan of the work to be done for FYP-2.

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[17] "Eightfold talent acquisition," 2024.

[18] "Ezyhire," 2024.

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Chapter A First Appendix if Required

If you want to add appendices then make sure you provide a proper title. Add different appendices by using the chapter command as shown for this one. They will automatically be added to the table of contents.

A.1 References

See how .bib file is prepared. The bibliography would be automatically added by Latex. So here is how you cite papers (you'll need to view the .bib file to understand). [**?**] is the correct way to make one citation and [**?** **?**] to make multiple citations. You can see various reference styles for manuals [**?**], conferences [**?**], websites [**?**] and journal articles [**?**]. Also, note that all citations appear in the order in which they appear within text. They are not sorted according to author names.

A.2 Equations

Make sure you use the math environment for writing equations. Again, make sure you do not hard code equation numbers. Use label and ref commands of Latex to number them and reference them. Here is an example of (A.1)

$$E = mc^2 \tag{A.1}$$

Note, how the equation is referenced by its equation number within brackets.

A.3 Figures

To insert a figure, you can use the following code as template. Here the figure is referenced using ref command of Latex. So Figure A.1 is the FAST logo. DO NOT HARD CODE FIGURE NUMBERS. Let Latex manage them for you.

Figure A.1: Fast Logo is the Caption.

A.4 Tables

Note how tables can be added in Latex. Also, make sure you use label and ref commands to define and use table numbers. You can use Table A.1 as a template to make tables, add captions to them and reference them. DO NOT HARD CODE TABLE NUMBERS. Let Latex manage their numbering for you.

If you are having problems in managing Latex tables, you may use Lyx https://www.lyx.org/.

Table A.1: Give a Caption to the Table.

This is where you will provide information about this table, any assumptions, or anything which a nontechnical reader must know to understand.

No.	Column 1	Column 2
1.	Some data	More data
2.	Some values	More values

A.5 Pseudo Code

When you need to explain an existing algorithm of an algorithm you devised, then you should use pseudo code notation. Given below, algorithm 1, is an example. Note how the numbering is managed by the Latex itself.

Data: this text

Result: how to write algorithm with LaTeX2e

initialization;

while *not at end of this document* **do**

  read current;

  if *understand* **then**

    go to next section;

    current section becomes this one;

  else

    go back to the beginning of current section;

  end

end

Algorithm 1: How to write algorithms

A.6 Code of Programming Languages

If you need to add C++, Java or any other programming language's code, then you can use "listings" package. Following is an example. Normally, the code is only added in the appendices to avoid clutter in the document.

```
#include <stdio.h>
#define N 10
/* Block
 * comment */

int main()
{
    int i;

    // Line comment.
    puts("Hello world!");

    for (i = 0; i < N; i++)
    {
        puts("LaTeX is also great for programmers!");
    }

    return 0;
}
```

A.7 Recommendations for LaTeX

Here we are providing you with only two Latex source files. main.tex and fypbib.bib file. You can also start a project at overleaf.com and upload the zip file of this template there and share the Latex source between your group and supervisor. This way all of you can work on the Latex files together online.